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Siluro-Devonian palaeomagnetism, terrane emplacement and rotation in the Caledonides of western Ireland

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SUMMARY

The palaeomagnetism of a Silurian succession (Lough Mask Formation lavas and sediments) and intrusive rocks (Kilbride dolerites and andesites and microgranodiorites, Killary Harbour area) in the Connemara–Mayo segment of the Irish Caledonides is described. Magnetizations are related to the regional tectonic history with the aid of fold, conglomerate and contact tests. They describe sequences of shallow to intermediate inclinations in both the eastern and western sectors of this Caledonian inlier. The shallow components appear to pre-date Siluro-Devonian folding while the intermediate ones post-date this episode. The change in palaeofield inclination is identical to the transition described by contemporaneous deep level (Silurian) to shallow and surface level (Lower Devonian) bodies in the British and Scandinavian Caledonides but declinations are rotated consistently to the west. The Lower–Middle Silurian vector is rotated clockwise by *c.* 100°. This difference is progressively reduced to *c.* 50° by the time of the Siluro-Devonian D2 folding, showing that this fold generation (now oriented E–W) developed in parallelism with contemporaneous folds on the NE strike continuation of the Caledonides. Post-folding remanences continue to deviate in a clockwise sense from the remainder of the Caledonides showing that rotation continued into Middle–Upper Devonian times and finally ceased during Carboniferous times. The integrated effect of this rotation is recognized in Ordovician magnetizations from the Dalradian metamorphic terrane of south Connemara. It can be explained by block rotations within a zone of distributed deformation according to the McKenzie–Jackson model and illustrated by many neovolcanic zones. Palaeomagnetism defines a tectonic regime commencing with rotations of *c.* 3° Myr^{-1} and lasting from mid-Silurian to Carboniferous times. Western Ireland is sited within a closure gap along the Iapetus suture extending from Ireland to Newfoundland, and these terranes were apparently emplaced by progressive elimination of this gap during sinistral transpression across the Caledonian orogen.

Key words: Caledonian terranes, Devonian, palaeomagnetism, Silurian, tectonic rotation, western Ireland.

1 INTRODUCTION

Caledonian terranes of western Ireland south of the Clew Bay Fault Zone (Fig. 2) form an inlier comprising two belts: a metamorphic belt of late Precambrian rocks (the Dalradian Metamorphic Terrane of Connemara, referred to hereafter as the DMT) is correlative with the Dalradian rocks in the Grampian Highlands of Scotland. This is juxtaposed against a thick, esentially unmetamorphosed, succession of Ordovician sediments comprising the South Mayo Trough (SMT, Fig. 2). The relationship between the two divisions is obscured by an unconformably overlapping Silurian (Upper Llandovery–Wenlock) succession. These two terranes occupy an anomalous position within the Caledonian orogenic belt (Phillips, Stillman & Murphy 1976; Leake 1986) for two reasons: (i) the structural grain is E–W to ESE–WNW in contrast to the familiar SW–NE grain characterizing the orogen throughout its continuation to the NE (Fig. 1); and (ii) the sedimentary troughs of Ordovician and Silurian age occur on the north side of the

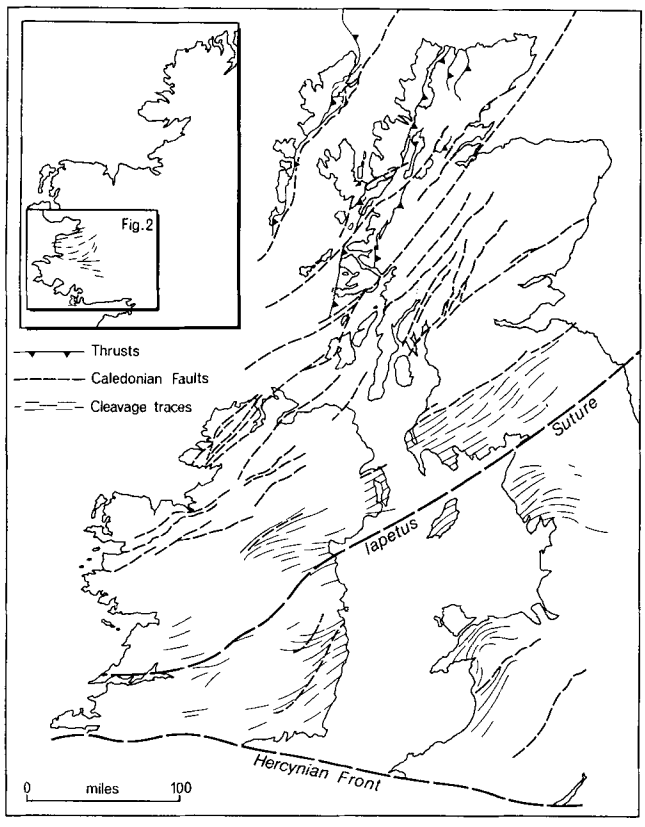


Figure 1. Late Caledonian structural trends in Connemara, western Ireland, contrasted with trends of comparable structures in the eastward continuation of the Caledonides in Britain and Ireland. The data are derived mostly from Soper & Hutton (1984) and the boxed area is the study location of Fig. 2.

metamorphic terrane; comparable troughs elsewhere occur exclusively on the south side of the metamorphic (orthotectonic) Caledonides.

Leake *et al.* (1983) evaluate these anomalous tectonic features and interpret the Mannin Thrust in south Connemara as the outcrop of a low-angle detachment carrying the Dalradian at least 10 km, and probably nearer to 50 km, southwards. The whole Dalradian section south of the SMT is regarded as an allochthonous body lying above this thrust emplaced between 500 and 380 Ma, with Rb–Sr data from mylonites indicating a mid-Ordovician (Llanvirnian) age of 460 ± 25 Myr. Following this climactic Caledonian deformation, transcurrent displacements occurred on major lineaments (see for example, Pitcher 1969; Phillips & Holland 1981; Hutton 1987) with a history of repeated sinistral displacement between mid-Ordovician and Devonian times (Hutton & Dewey 1986). Motions are well illustrated, for example, by the Main Donegal Granite shear zone where emplacement of the pluton at *c.* 400 Ma occurred within the zone, and movements continued as the pluton consolidated (Halliday, Aftalion & Leake 1980; Hutton 1987).

Numerous studies of Tertiary to Recent orogenic zones have shown that palaeomagnetism can provide critical data for identifying allochthonous terranes and delimiting their kinematic history. Magnetic directions are of most value if field tests can establish their ages relative to tectonic events. Pre-emplacement remanences then constrain displacements with respect to the adjoining cratonic forelands, while the history of magnetic overprinting is of value as an indicator of (a) tectonic and/or thermal events related to terrane docking, (b) thermoviscous remanence (Kent & Miller 1985) or (c) thermochemical effects linked to fluid expulsion during orogenic loading (Oliver 1986).

Few studies have used palaeomagnetism to unravel the tectonic history of pre-Tertiary terranes. However investigations to date suggest that the Caledonides of western Ireland are a promising area for application. Deutsch & Somayajulu (1970) first identified aberrant Ordovician declinations in the SMT. Subsequent studies resolved a remanence record within the DMT (Morris *et al.* 1973; Morris & Tanner 1977; Robertson 1988) and the SMT (Morris *et al.* 1973; Morris 1976; Deutsch & Storetvedt 1988; Smethurst & Briden 1988). The results are only in part compatible with data from the strike continuation of the Caledonides in Britain, and a range of tectonic interpretations have been proposed.

The present study covers the lower part of the Silurian succession lying unconformably on Dalradian rocks near the eastern margin of the outcrop. This is stratigraphically below, and along the strike from, the Salrock Formation investigated by Morris (1973) and Smethurst & Briden

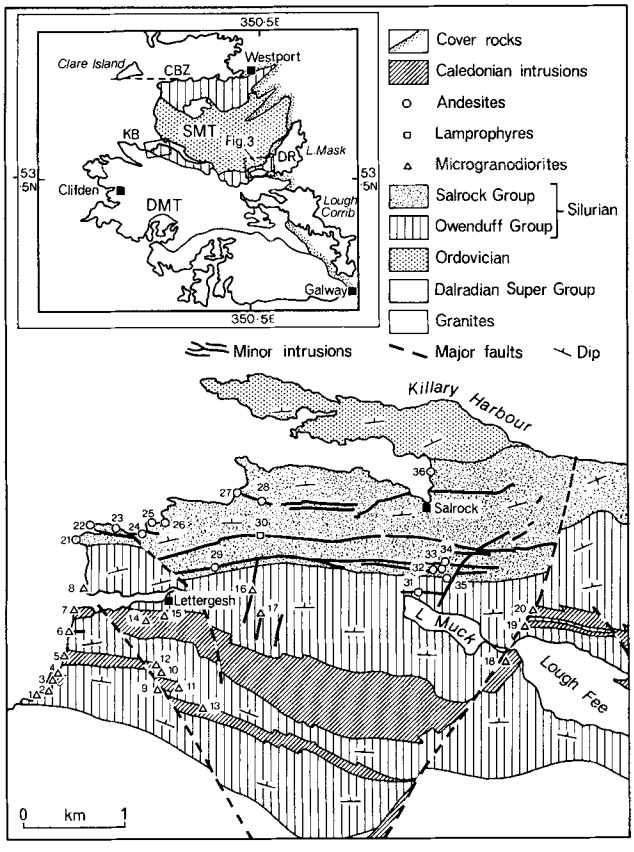


Figure 2. Simplified geology of the south western part of the South Mayo Trough, Connemara, with palaeomagnetic sampling sites in the Killary Bay area. The inset shows the areal distribution of the Dalradian metamorphic terrane (DMT) and the South Mayo Trough (SMT). Major fault zones referred to in the discussion are: CBZ (Clew Bay Fault Zone) and KB–DR (Killary Bay–Doon Rock/Loch Nafooey).

(1988). In addition igneous units are investigated comprising dykes and sills intruding the Silurian and Dalradian rocks spanning the exposed width of the inlier between Killary Bay in the west and the Kilbride Peninsula bordering Lough Mask in the east. These bodies range from mid-Silurian to mid-Carboniferous in age (Mitchell & Mohr 1987) and are cut by sparse younger dykes belonging to the Tertiary igneous province (Mohr 1982). The collective data are integrated here with the earlier palaeomagnetic results to clarify Siluro-Devonian apparent polar wander (APW) and derive a model of progressive deformation within this sector of the Caledonian orogenic belt.

2 GEOLOGICAL OUTLINE

2.1 Llandovery sediments and lavas

Late Ordovician deformation in the South Mayo Trough (Phillips & Holland 1981) resulted in a major unconformity between Ordovician and Silurian successions. The latter comprises the Owenduff Group (Fig. 2), an Upper Llandovery to Lower Wenlock clastic succession reaching a maximum thickness of 1500 m, and the Salrock Group (Wenlock) comprising some 1000 m of red and green sandstones and siltstones (McKerrow & Campbell 1960; Piper 1972; Holland 1981). In the (eastern) Kilbride area (Fig. 3) a transgressive terrestrial to marine sequence (Gardiner & Reynolds 1912) commences with a clastic succession 600 m in thickness attributed to the Telychian, (the uppermost stage of the Llandovery) resting on an irregular surface of folded Ordovician rocks. An impersistent basal member, comprising breccias and conglomerates, is succeeded by massive cross-stratified red sandstones interbedded with siltstones and mudstones; the sandstones and pebble beds were probably formed in a braided fluviatile environment (Holland 1981) with siltstones and mudstones recording a lower energy playa environment. Collectively these rocks, grouped as the Lough Mask Formation by Piper (1972), reach a maximum thickness of 170 m and are sampled in detail at four locations [Sites 8 to 11 in Fig. 3(a)].

Following deposition of impersistent basal conglomerates

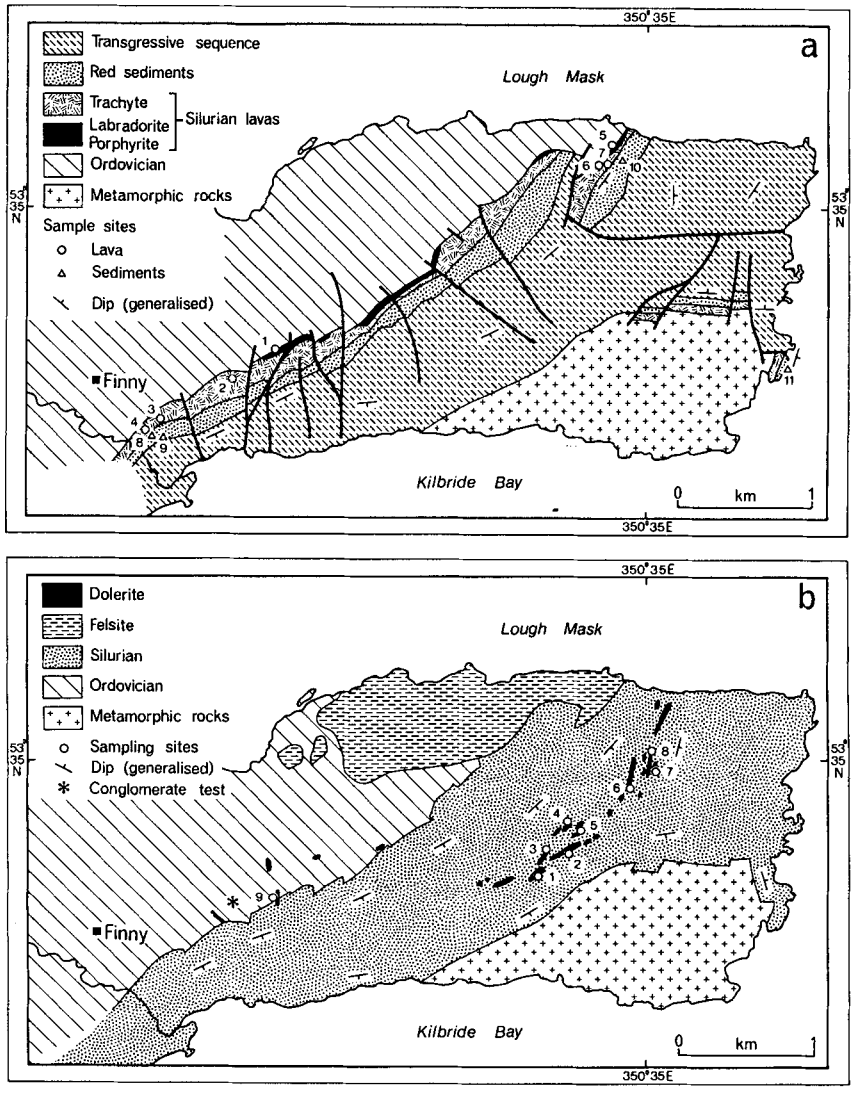


Figure 3. Simplified geology based on Gardiner & Reynolds (1912) and palaeomagnetic sampling sites in (a) the Lough Mask Formation and (b) the intrusive dolerite suites of the Kilbride Peninsula.

and breccias, albite trachyte (keratophyre) lavas were erupted prior to deposition of the remaining members of the Lough Mask Formation. The basal lava is feldspar-phyric and mapped as a labradorite porphyrite by Gardiner & Reynolds (1912); they noted abundant interstitial magnetite and haematite regarded as secondary. In succeeding aphyric lavas all ferromagnesian minerals appear to have been replaced by secondary phases. These rocks were folded in post Wenlock to early Devonian times to form the south easterly limb of a plunging E–W anticline. Sites were sampled on the south (1–4) and east (5–7, Fig. 3) limbs to link remanence acquisition to the timing of this folding. Outcrop extends sporadically 25 km to the west and the aphyric lava member is exposed on the south side of Lough Fee where a further site (12), in this case dipping to the north east, was also sampled at 350° 8′E, 53° 34′N.

2.2 Intrusive episodes

Several suites of Caledonian minor intrusions are emplaced within the Silurian outcrop occupying the strip of country extending some 35 km from Gowlaun on Killary Bay to the Kilbride Peninsula (Figs. 2 and 3). They also extend sporadically into the metamorphic terrane to the south (Stillman 1981) but are not found in Ordovician rocks overthrust from the north; magma ascent was concentrated along the tectonic boundary between these terranes (McKerrow & Campbell 1960). With the exception of sparse dolerite dykes which are largely unaltered and linked on structural, petrographic (Mohr 1982), radiometric (Mitchell & Mohr 1987) and palaeomagnetic (P. Dagley, personal communication) evidence to later events, these intrusions are altered with ubiquitous development of secondary chlorite, calcite and quartz. They were mostly emplaced prior to folding and faulting, and before full dewatering of the host sediments (McKerrow & Campbell 1960). According to McKerrow & Campbell they include dykes and sills of lamprophyre, microgranodiorite, andesite and felsite. These divisions are retained for purposes of this paper although in need of revision: some 'andesites' are comparable to the dolerites in the Kilbride area and some 'lamprophyres' represent autometasomatized and volatile-rich dolerites (P. Mohr, personal communication).

The microgranite suite reaches its most spectacular development in the Binn Chuanna Sill (sites 14 and 15 in Fig. 2) which is probably post-tectonic and a satellite intrusion of the Galway Granite emplaced into the DMT (Fig. 2 inset and P. Mohr, personal communication). They occur also as dykes extending upwards and downwards from the sills and are not observed to the east of Lough Fee. They are sampled at 20 locations (Fig. 2). The andesites are more widespread as dykes and sills throughout the Silurian outcrop and the northern extremity of the Dalradian outcrop to the south. Although commonly porphyritic in augite, this mineral and the groundmass are typically altered to secondary assemblages. They are sampled at 15 sites (Fig. 2) with two additional sites from the shore of Lough Corrib at Site 38 (351° 36′E, 53° 20′N) and site 39 (351° 36′E, 53° 31′N, see McKerrow & Campbell 1960). Intrusions of lamprophyre are widespread but profoundly altered and only two sites are included in this collection: one (30, Fig. 3) is from Killary Bay and the other (Site 37) is a thick dyke near the former Clonbrone School on the Dooros Peninsula (351° 33′E, 53° 29′N).

In the Kilbride Peninsula sills of dolerite are emplaced into the transgressive marine sequence. Primary augite and plagioclase are variably preserved, but chlorite and calcite are common alteration products. They are sampled at seven localities (Fig, 3b) on the south and east limbs of the plunging anticline folded by N–S compression during later stages of the Caledonian Orogeny in Upper Silurian (post-Wenlock) and early Devonian times (Phillips & Holland 1981).

Sparse Carboniferous dykes cut the Caledonian rocks in Connemara. They are sampled at Site 9 (Fig, 3b), which is probably an outlier of the TD dyke of Mitchell & Mohr (1987) dated 256 Ma (K–Ar), Site 10 and Site 40. The latter is the north-trending Logmor dyke of Mitchell and Mohr sampled at 350° 10′E, 53° 37′N in Killary Harbour just to the east of the area shown in Fig. 2 and dated 299 and 286 Ma by the K–Ar method. Mitchell & Mohr regard these dykes as a mid-Carboniferous intrusive episode overprinted by Variscan events at *c.* 300, 245 and 205 Ma.

3 FIELD AND LABORATORY METHODS

Cores were drilled in the field by portable two-stroke motor (Doell & Cox 1967) and orientated by sights on the Sun and/or magnetic bearings in conjunction with topographic sights. Cores were sliced into 2.3 cm long cylinders and magnetic remanences measured by parastatic magnetometers capable of reproducable measurements down to 0.01×10^{-5} A m^2 kg^{-1}. Approximately one half of the lava collection was progressively demagnetized by the alternating field (AF) method employing 500 cycle s^{-1} demagnetizers capable of peak fields of up to 140 mT; the remainder was progressively thermally demagnetized employing a Schonsted SM-2D thermal demagnetizer in steps of 100 °C to 500 °C and then in steps of 20 °C to between 600° and 700 °C. 75 per cent of the sedimentary sample, comprising red sandstones and siltstones, was thermally treated in similar steps to the lava collection, and the remaining 25 per cent was soaked in 2.5 N hydrochloric acid and measured at intervals over a period of 18 months.

Components constituting the total natural remanent magnetization (NRM) are identified from orthogonal projections with three successive steps of treatment taken as a minimum criterion for acceptability. Components common to two or more cores from the same sampling site are listed as group means in the data tables.

4 PALAEOMAGNETIC RESULTS

4.1 Albite trachyte lavas, Kilbride Peninsula

These lavas exhibit both deuteric and hydrothermal alteration and remanence composition is correspondingly complex both within and between sites. Components are partially magnetite-resident and subtractable by both AF and thermal cleaning (Fig. 4). Dual component behaviour is usually recognized in the magnetite range: samples 5-13 and 6-11 are examples of SE-directed positive components removed to recover a higher blocking temperature westerly component. Elsewhere haematite-held components have the

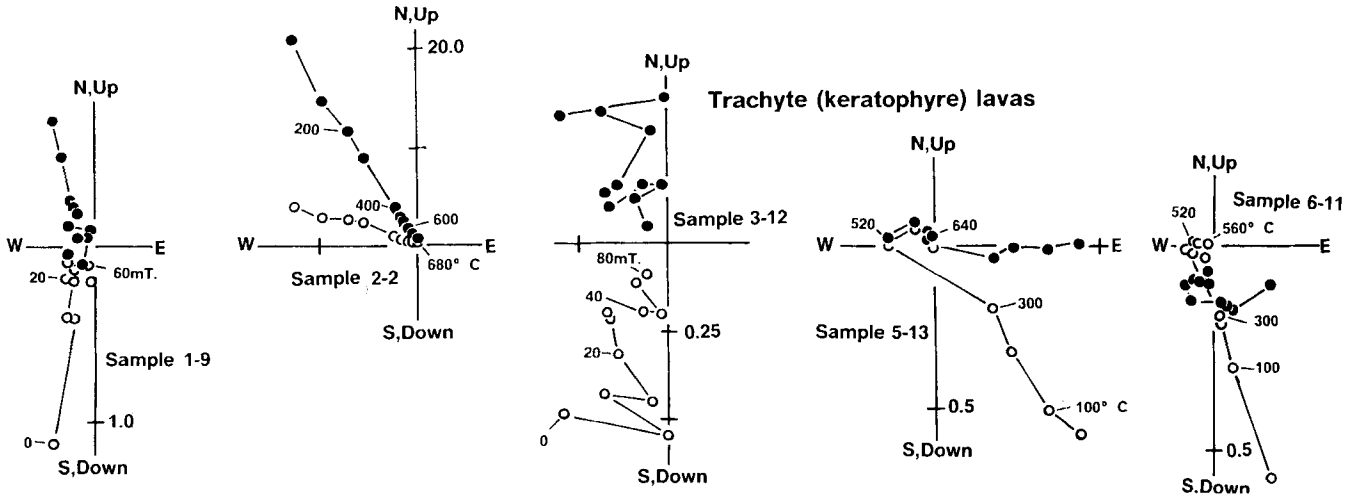


Figure 4. Orthogonal projections illustrating behaviour of samples from the trachyte lavas with progressive AF and thermal demagnetization. The closed symbols are projections onto the horizontal plane, the open symbols are projections onto the vertical plane and directions are *in situ*. Axes are scaled in units of intensity $\times 10^{-5}$A m^2 kg^{-1} and the AF demagnetizing fields are in milliTeslas (mT).

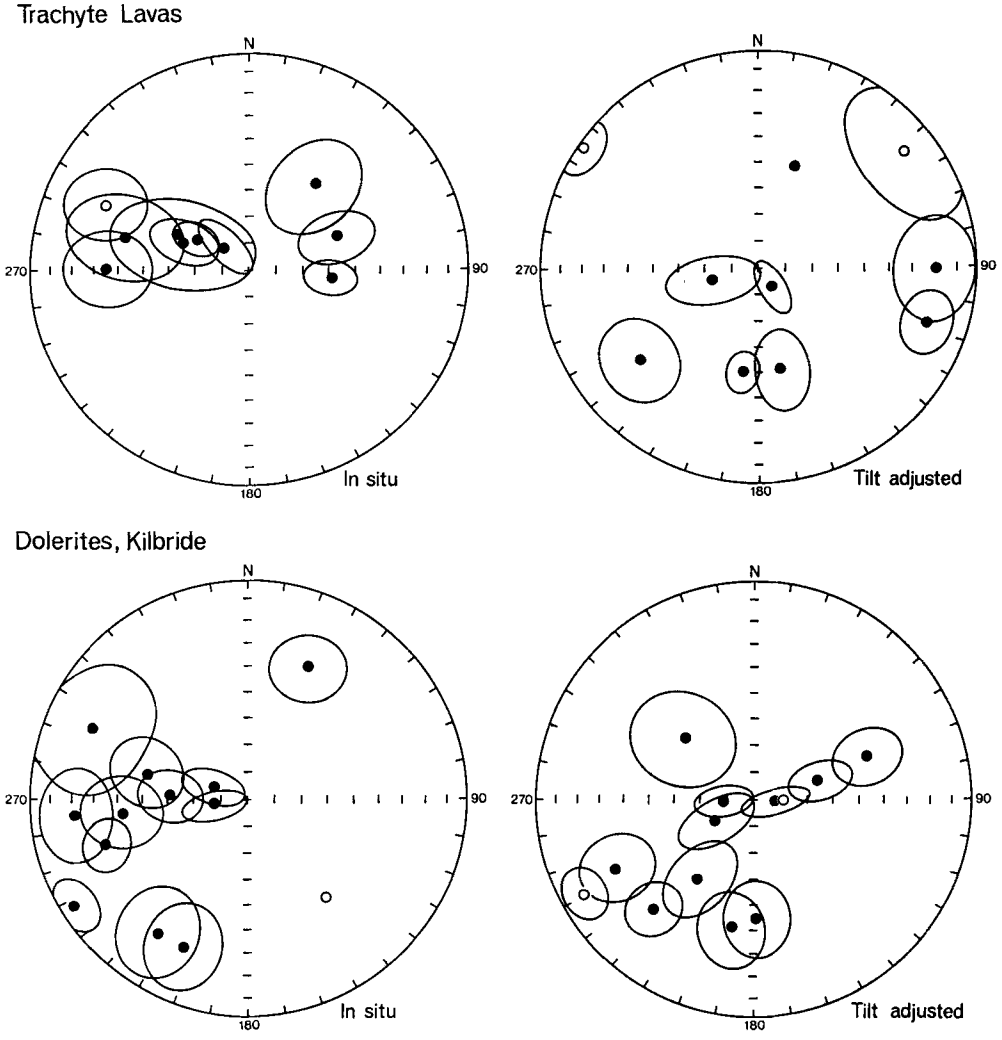


Figure 5. Site mean directions of magnetization from the trachyte lavas and Kilbride dolerites with 95 per cent confidence circles before and after adjustment for geological tilt. Pole-centred equal area projections.

same direction as magnetite components (e.g. 2–2) and this is therefore likely to be of primary deuteric origin. Samples 1-9 and 3-12 are examples of the W to NW positive components prominent in this collection.

In situ easterly and westerly directed magnetizations with shallow to intermediate positive inclinations are represented at site level (Fig. 5). The W positive group is randomized when adjusted for tilt: the ratio k_1/k_2 (where k_1 and k_2 are the precision parameters before and after adjustment for geological tilt) is 4.2. This identifies a negative fold test at the 95 per cent confidence level according to the simple test of McElhinny (1964); the more lenient test of McFadden & Jones (1981) is not applicable here because these sills occur on only one limb of the major structure. The predominant remanence ($D = 290°$, $I = 55°$, $N = 6$ sites, Table 1) is therefore secondary and syn- or post-folding in origin. Components identified from four sites (2, 5, 6 and 8) fail to fall within this westerly positive group in the *in situ* orientation and, when tilt adjusted, components from 5 and 6 are nearly antiparallel to the component at Site 2. The ratio of k_1/k_2 is 5.7: this in not quite significant at the 95 per cent confidence level of McElhinny's test owing to the small number of sites involved, but is suggestive of a pre-folding origin (Table 1) compatible with the probable deuteric

Table 1. Palaeomagnetic results, Albite trachyte lavas, Lough Mask Formation, Connacht.

Site	N(n)	R	α_{95}	In Situ D	In Situ I	Tilt Corrected D	Tilt Corrected I
1	10(4)	3.90	17	314	77	169	49
2	8(5)*	4.94	9	295	-25	305	-2
3	16(11)	10.64	9	306	65	188	47
4	10(9)	8.18	17	272	31	233	29
5	13(7)*	6.83	10	95	54	107	17
	13(4)	3.84	22	286	36	262	71
6	11(4)*	3.89	18	69	50	89	15
7	8(8)	7.49	15	299	56	154	83
8	11(6)	5.53	21	38	44	51	-12
	11(3)	2.89	29	294	61	20	44

	D	I	N(n)	R	k	α_{95}	Palaeomagnetic Pole $^{\circ}E$	$^{\circ}N$	dp/dm
(i) Shallow E-W axis (sites 2, 5 & 6):									
Tilt Corrected:	287.4	-11.7	3	2.88	17	31.3	243.0	5.3	18.4/36.5
In situ:	275.8	29.8	3	2.31	3	-	-	-	-
(ii) W Intermediate Group:									
Tilt Corrected:	207.5	72.1	6	4.80	4	36.5	334.7	23.0	56.9/64.5
In situ:	289.9	55.3	6	5.72	18	16.3	269.9	39.4	16.6/23.3
(iii) S Shallow Group (Variscan Overprints);									
In Situ:	182.6	-3.9	(9)	7.90	7	20.6	347.2	-37.9	10.3/20.6

FOOTNOTE: R is the magnitude of the resultant vector derived from components residing in n samples in a total site population of N. The components are recognised by linear segments on orthogonal vector plots and defined by minimum criteria of three successive demagnetisation steps yielding α_{95} <20° with precision k = (N-1)/(N-R) and cone of 95 percent confidence about the mean direction of α_{95}. D is the mean declination (°E) and I the mean inclination (positive below the horizon, negative above) and dp and dm are the semi axes of the 95 percent confidence oval about the palaeomagnetic pole in the colatitude direction and at right angles to it respectively. * Components interpreted as primary.

nature of the remanence at Site 2. Lower blocking temperature (5-13, 6-11) components and sample 2–2 illustrated in Fig. 4 are examples defining this axis. They identify a N→R inversion of the ancient field in this succession, the lower feldspar-phyric lava [Sites 5 and 6 in Fig. 3(a)] being of normal polarity; the single aberrant direction at site 8 may then record a transitional field.

4.2 Lough Mask Formation redbeds

Strong red to purple colouration in this unit is indicative of haematite, and AF demagnetization proves to be largely ineffective. Chemical and thermal demagnetization are more successful. They isolate a complex history of remanence acquisition both within, and between, the four sampled successions (Fig. 3a) reflecting complex lithification and diagenesis (Section 2.1). With progressive thermal cleaning (Fig. 6) an easterly to southeasterly-directed component with shallow inclination is most commonly recovered and present in the higher blocking temperature fraction close to the Curie point of specular haematite (Fig. 6). A more westerly component with shallow inclination is sometimes first subtracted in the lower blocking temperature range (samples 8-6 and 9-2 in Fig. 6); a low blocking temperature (<400 °C) W–SW positive component (samples 8-6 in Fig. 6) is also a minor feature of this collection. Southerly directed components with shallow inclination are present in other samples (8-12 in Fig. 6) and also occur as a higher blocking temperature component in some samples in which the easterly shallow component is present in the lower blocking temperature fraction (sample 8-16).

This southerly shallow to intermediate negative ('K') component is resistant to progressive acid leaching (Fig. 7). The form of J/Jo against leaching time suggests a roughly constant resistance due to slow solution of large authigenic overgrowths (e.g. Turner 1980). Behaviour of samples with easterly to south easterly directions (the 'J' group, see Fig. 8) is variable but they are generally more readily leached. This is shown by samples 5 and 10 from Site 8 in Fig. 7, where no vestige of the southerly shallow component is apparent, and by samples 5 and 9 from Site 10 where the SE component appears to be present in two phases. Full recovery of magnetizations progressively acquired by authigenic haematite growth in a reversing field is illustrated in sample 15 at Site 8 (Fig. 7). The sample 9 direction is similar to the overprint in the underlying lavas but is otherwise unrepresented except as the low blocking temperature component in a few thermally treated cores (8-6 in Fig. 6). The steep fall in J/Jo with time (e.g. 5 and 9 from Site 10) is more likely to represent subtraction of fine pigmentary haematite; the ensuing slow decline may record a carrier resident in large specularite (primary) or authigenic (secondary) grains because their relative surface areas would gradually increase as leaching proceeds.

The variety of results from chemical leaching experiments indicate a complex paragenesis of haematite in this formation, while the lower blocking temperature components evident in thermal treatment imply that detrital magnetite may also be a remanence carrier. The bulk of the samples have E–SE shallow ('J') directions; they are correspond approximately with the pre-folding field axis recorded in the underlying lavas. No single population of

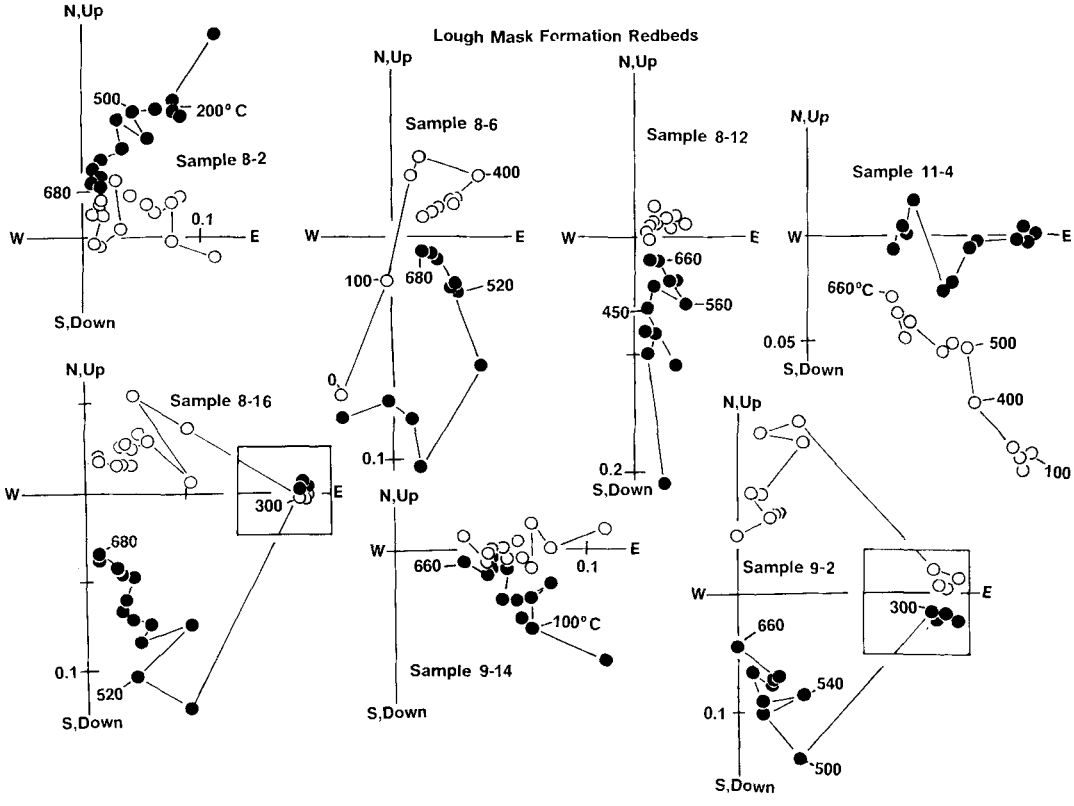


Figure 6. Orthogonal projections illustrating behaviour of samples from the Lough Mask redbeds with progressive thermal cleaning. Symbols are as for Fig. 4.

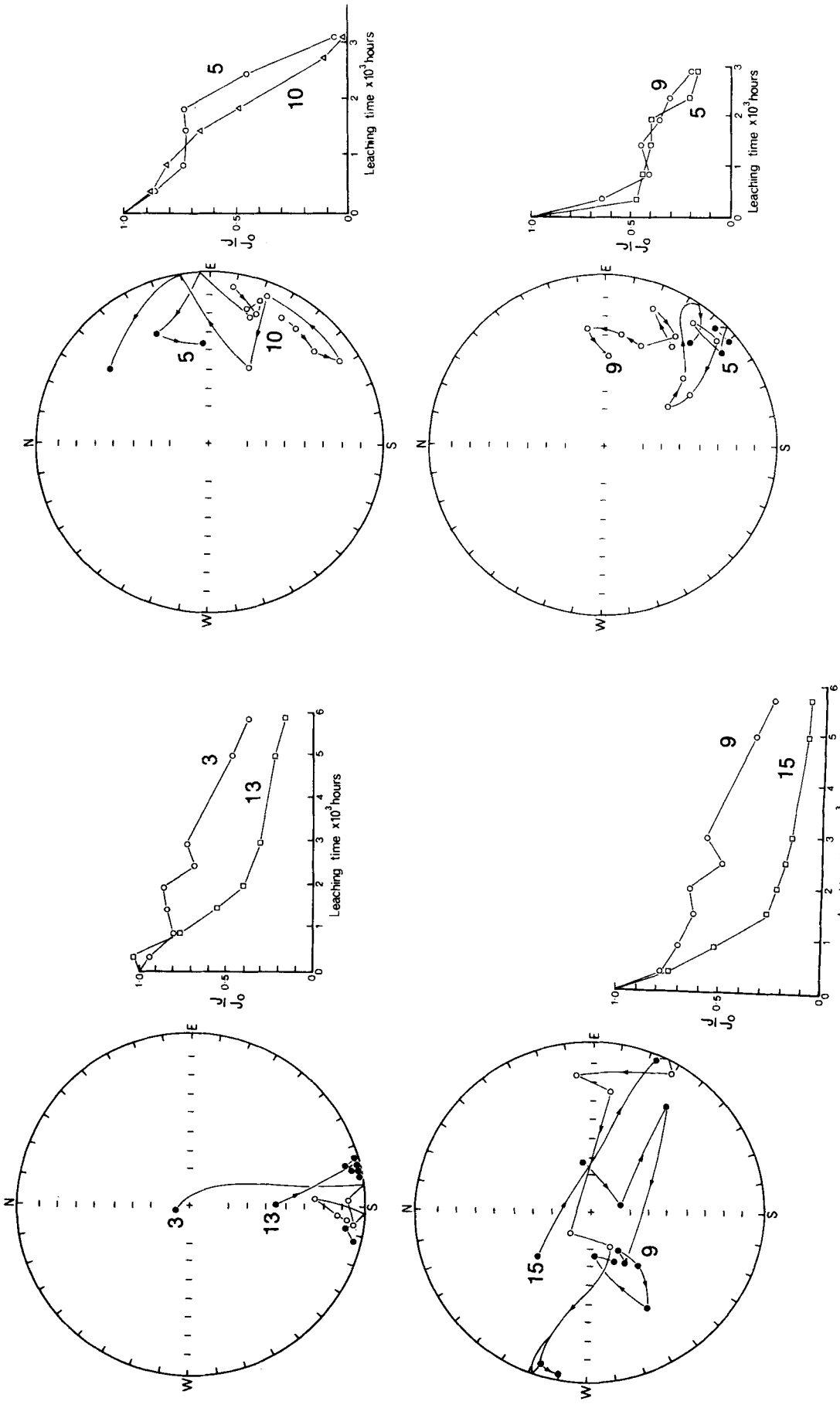


Figure 7. Behaviour of *in situ* direction and intensity of magnetic remanence in samples of the Lough Mask redbeds with progressive leaching by hydrochloric acid. Paired samples are from Site 8 (5 and 10), Site 11 (3 and 13), Site 10 (9 and 5) and Site 8 (9 and 15). See text for explanation.

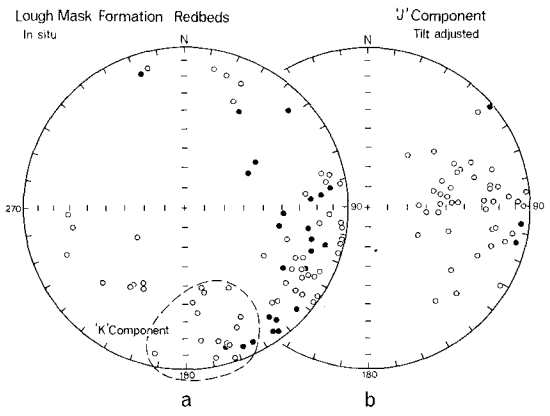


Figure 8. Components of magnetization isolated by thermal and chemical demagnetization from the Lough Mask redbeds. Low blocking temperature components (<300 °C) are excluded from this figure.

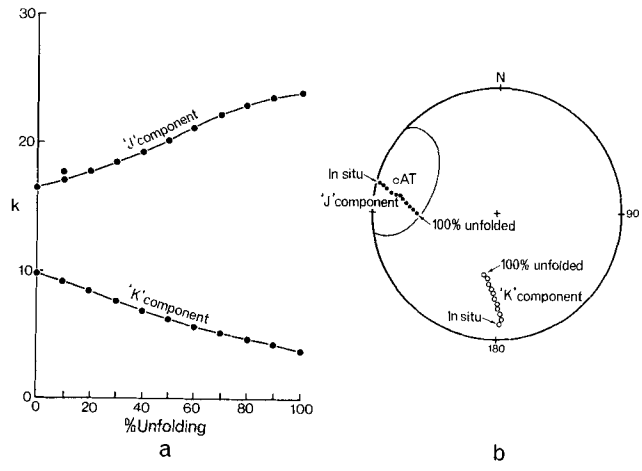


Figure 9. (a) Change in the Fisher precision parameter, k, for the 'J' and 'K' components in the Lough Mask redbeds as calculated from the four sites during progressive unfolding of the structural tilt. (b) Change in mean direction of remanence of the 'J' and 'K' components during this unfolding. AT is the mean direction and 95 per cent confidence circle for the three sites in the albite trachytes inferred to carry a probable primary remanence (Table 1).

components is present however, and there is a smearing towards southerly shallow ('K') directions more typical of the Upper Palaeozoic field direction (Fig. 8) in this area.

The 'J' components occur at all sites but inclinations are variable: at 8 and 9, where the geological tilts are southerly, inclinations tend to be negative, whilst at Sites 10 and 11 where tilts are easterly some single components with this declination have positive inclinations (*cf.* 5, 9 and 10 from Site 8 and 5 from Site 10 in Fig. 7, and 11-4 in Fig. 6). Hence these components appear to pre-date some or all of the regional folding. It is recognized (e.g. Miller & Kent 1986) that effects such as fluid migration associated with deformational events are potent agents of remanence acquisition in red beds. To test this possibility site mean directions of the 'J' and 'K' groups (arbitrarily separated by $D = 140°$ to yield approximately circular distributions) are analysed in Fig. 9 at each 10 per cent stage of unfolding. Precision of the group mean improves continuously as the beds are unfolded in the case of the 'J' component, and deteriorates continuously as beds are unfolded in the case of the 'K' component. Hence acquisition of the 'J' component appears to have been essentially complete by Siluro-Devonian folding of these beds, whilst the 'K' component clearly post-dates this folding.

Although the 'J' component is closest to the probable primary axis identified in the underlying trachyte lavas in the *in situ* orientation (Fig. 9b), this is fortuitous because the lava axis is only recognized by correction for tilt while the progressive adjustment of the 'J' component for tilt implies that it too, is pre-folding. Hence the results imply movement of the region into higher palaeolatitudes between acquisition of the primary component by the lavas ($\lambda = 6°N$) by a thermal or thermo-chemical mechanism in late Lower Silurian times, and acquisition of the 'J' component in the overlying sediments ($\lambda = 16°S$) by a post-depositional mechanism but prior to folding in Upper Silurian–Lower Devonian times.

The *in situ* direction of the 'K' component ($D/I =$ 173°/14° calculated at sample level, see Table 2) plots closest to the palaeofield direction predicted for this area from the mean APW path for Eurasia at 270–260 Ma (Irving & Irving 1982; Piper 1987). It is therefore regarded as a late Variscan overprint. In Ordovician rocks *c.* 10 km to the north Deutsch & Storetvedt (1988) identify an overprint of $D/I = 182°/+6°$ (*cf.* Fig. 8b) corresponding to the regional field at 300–290 Ma, while 40 km to the west Smethurst & Briden (1988) identify a prominent overprint (classed as 'I') with a mean $D/I = 195°/14°$ corresponding to the regional field at 320–310 Ma. A sporadic overprint with similar direction ($D/I = 183°/-4°$) is also present in some samples from the trachytes (Table 1). Hence there is widespread evidence (next section and Table 4) for overprinting in this area during a broad time interval embracing the latter part of the Variscan Orogeny. This is recognized as an isotopic event by Mitchell & Mohr (1987).

4.3 The Kilbride dolerites and assessment of overprinting events

With the exception of a dyke (Site 9 of this study), the Kilbride dolerites are emplaced as sills predominantly within well-bedded calcareous sandstones of the Finny School Member of the Silurian succession at Kilbride; they are less common at higher stratigraphic levels (Gardiner & Reynolds 1912). Thermal cleaning of samples from nine sites (Fig. 3b) yielded no interpretable results, apparently because the NRM, is dominated by phase(s) readily altered by heating in air. Evidence for this is apparent from AF demagnetized results (Fig. 10) where large low-coercivity components are subtracted in the first step of treatment (Sites 6 and 8 in Fig. 10). Once this is removed, the remaining few per cent of the remanence comprises a component of rather poor stability with a westerly positive direction (Sites 6, 7 and 8 in Fig. 10).

This component is recognized in seven of the eight sills (Table 3). The *in situ* mean is $D/I = 269°/47°$ ($\alpha_{95} = 18°$, Table 4, Section 3) and the only coherent group of directions to emerge from treatment of this collection. Grouping deteriorates with adjustment for tilt of the

Table 2. Red sediments, Lough Mask Formation, Connacht, Ireland.

	D	I	N(n)	R	k	α_{95}	Palaeomagnetic Pole ^{o}E	^{o}N	dp/dm
(i) 'J' Component									
Tilt Adjusted:	87.3	-31.9	(47)	41.5	8.4	8	93.0	-21.3	4.2/8.3
In Situ:	104.0	-2.6	(47)	40.7	7.3	8	69.9	-9.3	4.8/8.6
(ii) 'K' Component									
Tilt Adjusted:	184.9	-57.2	(35)	25.2	3.7	15	336.5	-73.8	15.7/21.6
In Situ:	173.1	-13.8	(35)	29.5	6.2	11	359.9	-43.0	5.6/10.9

FOOTNOTE: Symbols as for Table 1

intruded beds (Fig. 5). Precision, k, declines from 11.8 to 5.6, a negative test but not quite significant at the 95 per cent confidence level according to the criterion of McElhinny (1964).

If the characteristic remanence in these rocks is primary, tilt adjustment implies that the intrusions post-date Upper Silurian–Lower Devonian folding in this area. This latter conclusion was reached by Gardiner & Reynolds (1912) who failed to find evidence for offset of outcrop along north to northwest trending faults linked to this deformation (Fig. 3). It now appears more probable that the sills were emplaced prior to folding (P. Mohr, personal communication) and hence the remanence is secondary. Close similarity to the overprint direction in the trachyte lavas (*cf.* Tables 1 and 4), suggests that the two units record a common remagnetization event.

The range of pre- and post-folding components recognized in the Kilbride area implies however, that no pervasive regional overprinting event is recorded here. The magnetite-resident overprint in the lavas, for example, is only sporadically observed as an overprint in the red sediments although a haematite-held overprint of Variscan age is present in this latter unit (Fig. 8 and Table 2). Igneous contacts are poorly exposed with the exception of dyke Site 9. This dyke has stronger remanence and higher coercivity (Fig. 10) then the dolerite sills. Following initial treatment, it exhibits a characteristic component with a mean $D/I = 240°/5°$ (Table 3) contrasting with the sills and indicative of a different age of magnetization (see Section 2.2). The intruded lava is dominated by components which, although somewhat steeper (Site 1, Table 1) than the typical secondary remanence in the trachytes, have similar declination. No other coherent grouping was recognized here although a component with similar direction to the dyke remanence is present in the highest blocking temperature fraction immediately adjacent to the dyke contact (e.g. Fig. 11c). The secondary remanence in the lavas therefore pre-dates (Carboniferous) emplacement of the dyke (Mitchell & Mohr 1987). This conclusion accords with the observation that it does not correspond with any post-Caledonian palaeofield in this area.

Unfortunately, inadequate clasts of trachyte lava were found in gravel bands within the overlying red beds to permit a palaeomagnetic conglomerate test. However, coarse volcanic breccias are developed within the Lower Ordovician (Arenig) volcanic succession immediately

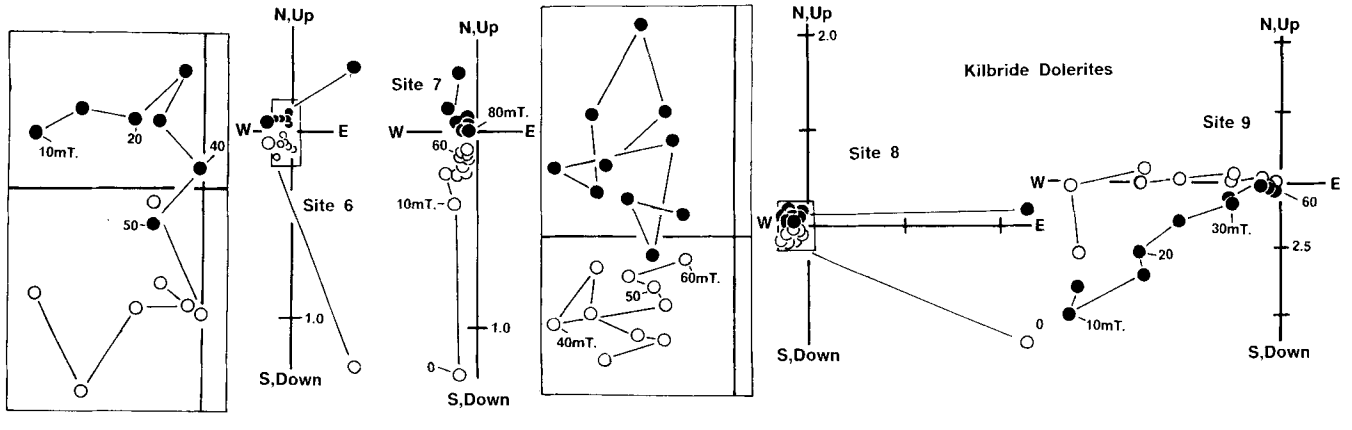


Figure 10. Orthogonal projections illustrating behaviour of members of the Kilbride dolerite suite with progressive AF demagnetization. Symbols are as for Fig. 4.

Table 3. Palaeomagnetic results from dolerite intrusions, Kilbride peninsula.

Site No.	N(n)	R	α_{95}	In Situ D	In Situ I	Tilt Corrected D	Tilt Corrected I
1	6(5)*	4.85	15	266	15	244	25
2	10(8)*	7.36	17	287	47	216	49
3	9(6)*	5.78	14	264	76	178	42
	9(2)	1.97	-	142	-37	81	-80
4	4(3)*	2.74	11	254	26	222	29
5	9(5)	4.92	15	25	31	68	40
6	9(3)	2.96	17	215	25	191	35
	9(5)*	4.78	18	268	36	242	74
7	10(5)*	4.82	17	274	56	71	84
8	9(4)	3.77	27	296	20	313	52
	9(2)*	1.98	-	283	70	78	69
9**	9(6)	5.91	9	240	5	242	-8

* Components used in group mean calculation (Table 4).

** Dyke cutting labradorite porphyrite lavas. Symbols as for Table 1.

Table 4. Summary of palaeomagnetic results, Caledonian intrusive suites, South Mayo Trough.

D	I	N	R	k	α_{95}	Palaeomagnetic Pole ^{o}E	^{o}N	dp/dm
1. MICROGRANODIORITES								
(i) W Shallow Group:								
In Situ:								
277.8	8.2	9	8.65	22.5	11	256.3	8.0	5.6/11.2
Tilt Corrected:								
282.2	0.3	9	8.60	19.8	12	250.3	7.3	5.9/11.9
(ii) W Intermediate Group:								
In Situ:								
265.7	55.2	7	6.68	18.8	14	286.1	25.7	14.4/20.3
Tilt Corrected:								
322.1	37.3	7	6.63	16.2	15	226.4	46.4	10.7/18.1
2. ANDESITES								
(i) W Shallow Group:								
In Situ:								
267.3	12.9	10	9.41	15.0	13	266.2	3.6	6.7/13.1
Tilt Corrected:								
282.3	14.4	10	9.60	22.5	10	254.5	13.2	5.5/10.6
(ii) W Intermediate Group:								
In Situ:								
278.2	54.5	8	7.71	24.4	11	277.0	32.1	11.4/16.1
Tilt Corrected :								
318.4	34.2	8	6.64	5.2	27	229.0	42.8	17.7/30.9
3. DOLERITES (Kilbride)								
(i) W Intermediate Group								
In Situ:								
269.0	47.3	7	6.49	11.8	18	279.1	21.9	15.4/23.8
Tilt Corrected:								
213.4	61.7	7	5.93	5.6	28	326.3	10.7	33.5/43.3
4. UPPER PALAEOZOIC OVERPRINTS:								
(Microgranodiorites, 5N13R)								
196.3	5.5	18	14.7	5.1	17	330.8	-32.0	8.6/17.2
(Andesites, 7N8R)								
191.6	-13.0	15	12.9	6.7	16	334.5	-42.0	8.3/16.2
(Dolerites, 3N10R)								
211.1	2.1	13	11.7	9.3	14	314.1	-29.6	7.2/14.4

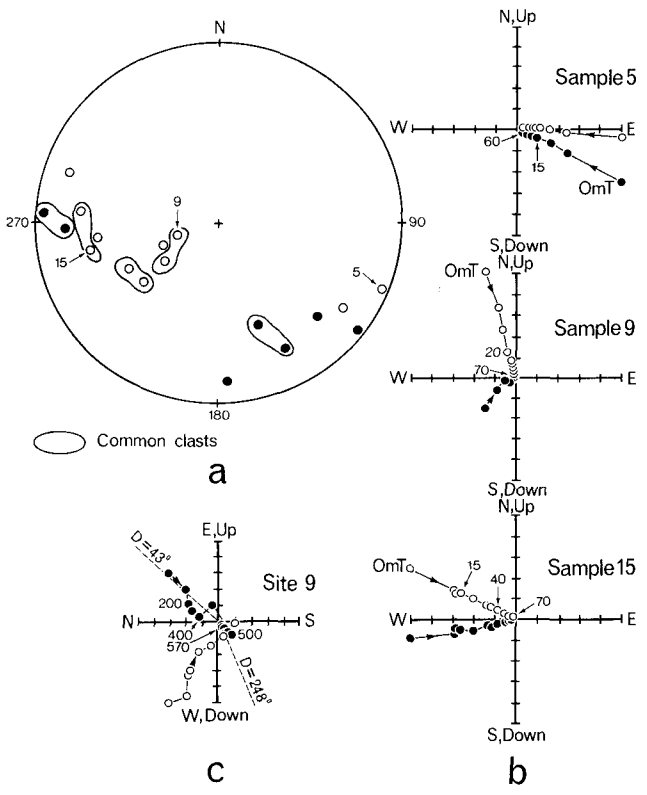


Figure 11. (a) Palaeomagnetic conglomerate test on clasts from a volcanic breccia underlying the Silurian succession in the Kilbride Peninsula: equal area projection of component mean directions. (b) Examples of the components defined by progressive demagnetization for samples illustrated in (a). (c) Dual component behaviour in a lava sample at the contact adjoining dolerite dyke Site 9.

underlying the Silurian outcrop (Fig. 3b). Ordovician rocks in this area are predominantly pillow lavas but include lenses of basaltic and silicic (felsite) breccias comprising vent agglomerates (Gardiner & Reynolds 1912, 1914). These deposits reach their most impressive development 1 km ENE of Finny close to trachyte lava sites 1 and 2, where blocks of felsitic composition similar to, but somewhat more silicic than, the trachytes attain diameters of up to 0.5 m; they are sampled from the location indicated in Fig. 3(b) and 10 cores are paired from common clasts. All 18 samples exhibit high magnetic stability and define only single components residing in magnetite [e.g. samples 5, 9 and 15 in Fig. 11(b)]. Directions are random between clasts but tight conformity is observed between cores from the same clast (Fig. 11a). Clearly no significant overprinting has influenced these rocks since the (Ordovician) eruption responsible for their formation. This is noteworthy both because this volcanic pile has been subjected to penecontemporaneous metasomatism leading to widespread development of chert, and because it was tightly folded prior to deposition of the Silurian rocks (the sample locality presently dips at 65° to the NW). In particular, there is no evidence for the W+ overprint component present in the overlying lavas.

Whilst partial overprinting of the red sediments can plausibly be linked to haematite growth during fluid migration motivated by the Variscan orogeny (*cf.* Fig. 17), the origin of the prominent secondary remanence in the trachyte lavas and Kilbride dolerites is less clear. A thermoviscous origin (Kent & Miller 1985) related to elevated temperatures during burial does not appear to be relevant to this example because (i) it has not influenced magnetite in the underlying breccias; (ii) it is not recorded in any mineral component in the redbeds; and (iii) it does not correspond to the Lower Carboniferous palaeofield direction applicable to the timing of deposition of platform successions bordering the inlier (Piper 1988). Hence it appears that hydrothermal fluid migration promoted by the Siluro-Devonian folding preferentially altered the permeable trachytes and susceptible mineral assemblages in the dolerites. This is recorded by their petrology (Section 2.1).

4.4 Intrusive suites, Killary Harbour area

A north positive component compatible with acquisition in the Cainozoic to Recent field is only sporadically represented in the andesites and removed by cleaning in low AF fields (e.g. 31-7 in Fig. 12); it may survive to 500 °C or above with thermal cleaning (22-7). It is comparable with the 'L' component much more widely recorded in the country rocks of this area by Smethurst & Briden (1988). Components resolved by cleaning are mostly characterized by westerly declinations; inclinations are distributed from shallow negative to steep positive and the population is clearly composite. The westerly shallow components are most discrete (28-1 in Fig. 12) but recovered in initial stages of treatment. Components with steeper inclinations tend to be more erratically defined (22-2, in Fig. 12) but are of higher coercivity and blocking temperature (31-7, 37-5 and *cf.* 28-1 and 29-3 in Fig. 12). There is also evidence for inversion of polarity in a few samples (23-4 in Fig. 12). Thermal treatment subtracts the W shallow component to recover a steeper westerly component at low temperatures and there is some evidence that the W shallow directions are of composite origin: a small residual westerly shallow component is again isolated above 500 °C in some samples. These complexities are illustrated in Fig. 13 by progressive thermal demagnetization results from paired cores at Sites 27 and 32. At Site 27 a SW+ component is retained in sample 3 to the Curie point of haematite within a distributed blocking temperature spectrum, whereas the W shallow component resides in magnetite with a narrow blocking temperature spectrum. At Site 32 the shallow westerly component is only recognizable within a narrow range of the blocking temperature spectrum in sample 7 close to the magnetite Curie point, and in sample 2 prior to isolating a haematite-resident example of the westerly positive population. Majority populations are apparent at most sites (Table 5), but both shallow and steep directions with contrasting properties are commonly recognized at the same site.

In the microgranite suite Cainozoic–Recent field components are again only sporadically represented; they are subtracted by low AF fields (14-3 in Fig. 14) but at temperatures in the higher blocking spectrum of magnetite (15-2). Westerly components subtractable by AF cleaning characterize most of this collection. The shallow W-directed directions again reside in lower coercivity components

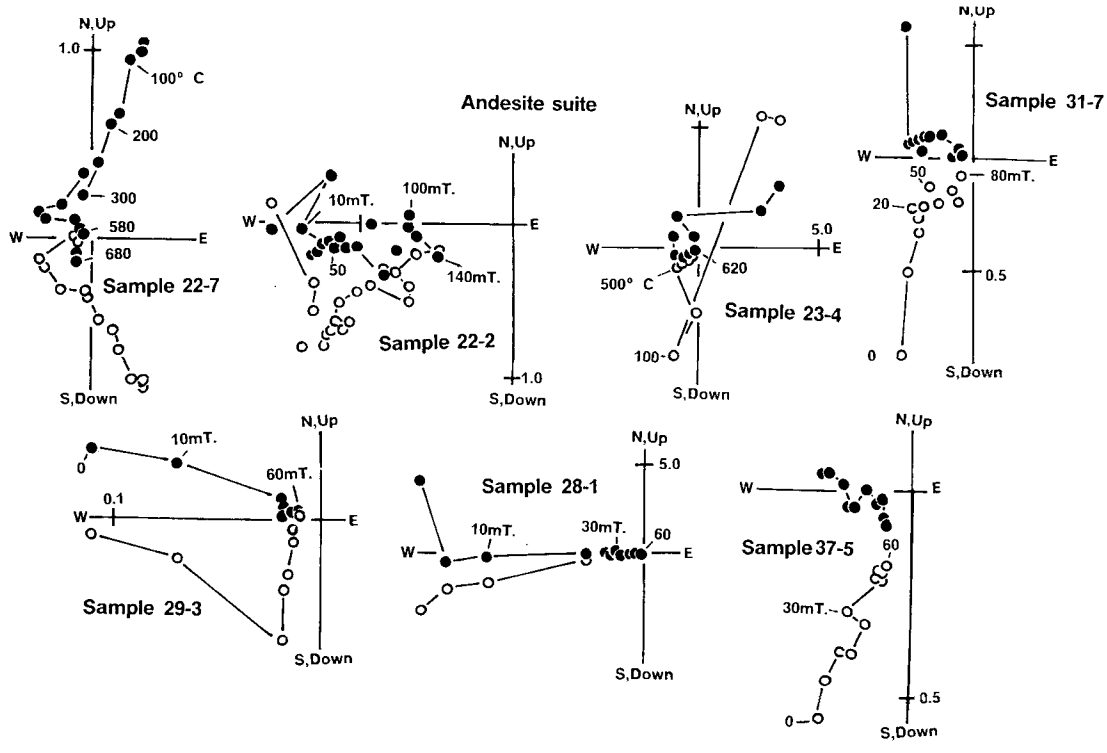


Figure 12. Orthogonal projections illustrating behaviours of members of the andesite suite with progressive AF and thermal demagnetization. Symbols are as for Fig. 4.

(10-10 in Fig. 14) than the steeper W components (14-3 and 19-5).

The distribution of westerly shallow negative to steep positive inclinations in these rocks was also recognized in earlier studies by Morris (1973, 1976) but, in the absence of detailed demagnetization, their complex origin was not then apparent. It contrasts with the remanence record in the host Salrock sediments (Smethurst & Briden 1988). Morris resolved directions of magnetization between $D = 226°–263°$ and $I = 4°–39°$ in country rocks at the contact zones which were in general conformity with remanence directions in the adjoining intrusions. Directions similar to components in adjoining intrusions were also identified in three contacts from this study (Fig. 15). Hence the contact tests imply that shallower inclinations in this westerly population are primary and record the palaeofield direction at the time of the igneous episode.

Westerly shallow to steep NRM components from thermal aureole zones are also shown by Smethurst & Briden in their Fig. 4. However, according to these authors a WSW negative direction and a more common antiparallel equivalent is observed in green sediments adjoining two of these intrusions; it appears to be controlled in some measure by proximity to the bodies [although an example illustrated (S10/2A) has a positive inclination like most of the results of the present study]. No significant results were recorded in the igneous units by Smethurst & Briden and there is no hint of the NE + /SW − axis identified by these authors in the remanence structure isolated in the intrusives of this study. Although this latter axis has been interpreted as a vestige of primary TRM in the intrusions, it is both largely antiparallel to the intrusives and has a direction [$D/I = 84°/26°$ (S10) and $69°/2°$ (S4)] closer to the mean characteristic remanence in the host Salrock Formation ($D/I = 56°/5°$). Hence it is probable that the remanence in these two contact zones is a later phenomenon acquired over a protracted period of time (reversals are present), and after the westerly directed remanence recognized in this study and by Morris (1973, 1976).

One microgranodiorite body follows a pre-existing NE–SW tear fault post-dating folding of the Silurian sediments, although it was apparently intruded before the sediments were fully dewatered (McKerrow & Campbell 1960). Some andesite dykes cut microgranodiorite sills and some members of both suites appear to post-date the folding. A palaeomagnetic fold test is not diagnostic because strike directions (Fig. 2) are close to remanence declinations. Site means (Fig. 15) show a prominent cluster of W shallow components and a more diffuse group of W positive components. Only one lamprophyre intrusion sampled (Site 37) yielded significant palaeomagnetic results (Table 5) which are considered with the remaining intrusions. At least two populations of directions appear to be present in Fig. 15, one with inclinations of less than 20° and the other with inclinations of more than 40°. Since there is a paucity of inclinations at *c.* 30° two representative means are calculated in Table 4 for each collection incorporating the westerly components with inclinations less than, and more than, 30° respectively.

The palaeomagnetic field was evidently largely of reversed polarity during the interval recorded by these magnetizations, although a normal field is recorded at Site 36 in the andesites and Site 9 in the microgranodiorites (Fig. 15). In each case the W+ group is dispersed by adjustment for tilt

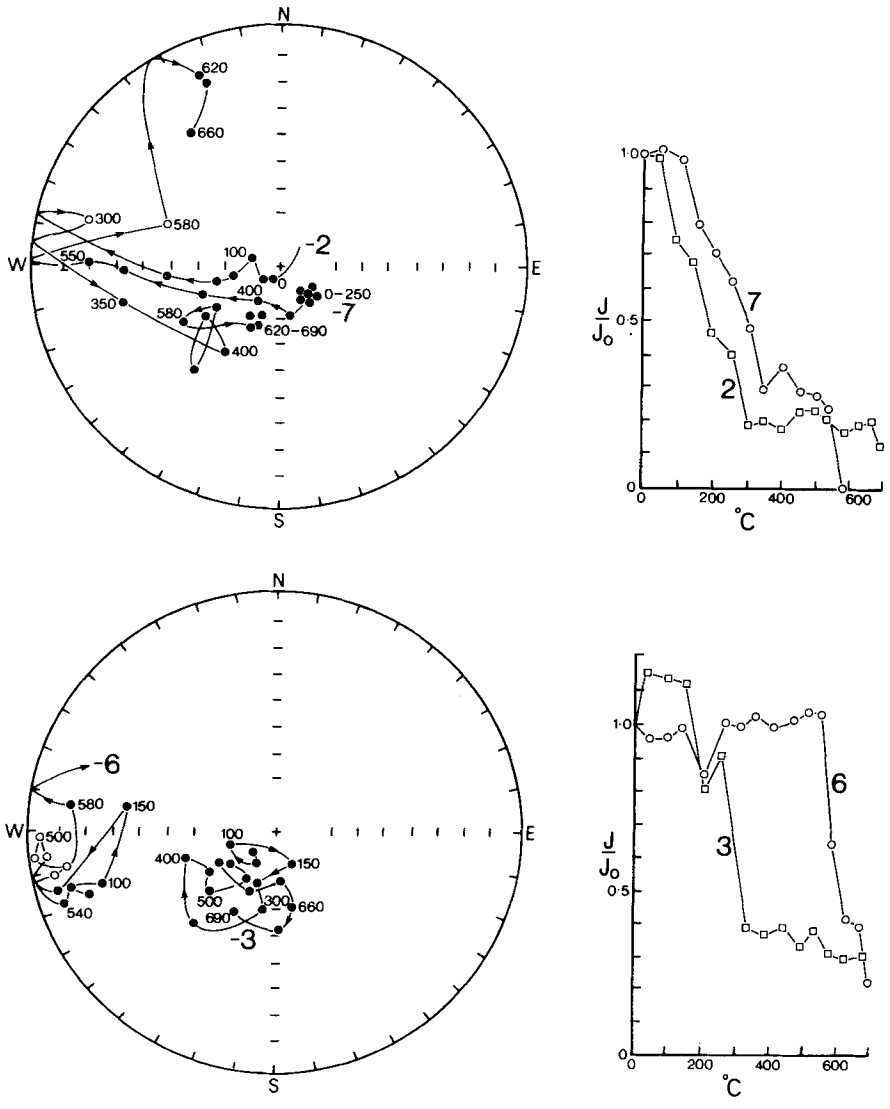


Figure 13. Contrasting *in situ* directional and unblocking temperature behaviour illustrated in pairs of cores from the same sites in the andesites showing both shallow and intermediate to steep westerly directions of magnetic remanence. Equal area projections.

Table 5. Palaeomagnetic results from andesite and lamprophyre intrusions, Killary area.

Site No.	N(n)	R	α_{95}	In Situ D	In Situ I	Tilt Corrected D	Tilt Corrected I
21	4(3)	2.91	26	283	21	295	-0
22	6(5)	4.79	18	252	20	282	30
23	7(4)	3.64	34	269	17	284	9
	7(2)	1.96	-	272	56	323	24
24	6(6)	5.74	16	327	71	350	14
25	7(3)	2.98	12	289	52	330	18
27	9(3)	2.96	17	266	9	277	12
	9(2)	1.99	15	259	57	323	37
28	10(4)	3.88	19	279	-18	274	10
29	13(13)	12.48	9	267	-1	273	15
31	8(8)	7.28	19	256	10	269	17
32	10(9)	8.86	7	268	4	274	8
33	7(5)	4.65	24	264	11	276	15
34	6(2)	1.99	-	298	60	335	24
35	8(5)	4.52	28	278	37	309	24
36	10(6)	5.77	15	320	-12	297	-55
	10(5)	4.86	15	40	-32	85	-46
37*	11(9)	8.68	10	287	61	287	61
38	9(9)	8.65	11	267	43	267	43
39	6(6)	5.80	14	258	43	258	43
40+	7(7)	6.86	9	201	4	205	54

* Lamprophyre dyke;
\+ Dolerite dyke, symbols as for Table 1.

of the intruded sediments (Table 4) and is clearly post-folding in origin. It includes the characteristic remanence at Sites 14 and 15 in the post-tectonic Binn Chuanna sill. The W shallow means are so close to the mean strike of the intruded beds that they are not significantly influenced by tilt adjustment; overall precisions nevertheless improve suggesting that these magnetizations record a remanence pre-dating the folding, a conclusion supported by data from the contact zones. The *in situ* W+ means from the andesite and microgranodiorite suites compare with means derived by Morris (1976) from andesites ($D/I = 257°/19°$, five sites), microgranodiorites ($D/I = 263°/53°$, nine sites) and lamprophyres ($D/I = 240°/38°$, eight sites); from evidence of thermal demagnetization in this study it would appear that remanences included in these groupings are of composite origin.

Younger dolerites emplaced between Caledonian times and the emplacement of the Lower Tertiary (Thulean, British Tertiary) igneous province are also represented in this area (P. Mohr, personal communication). The

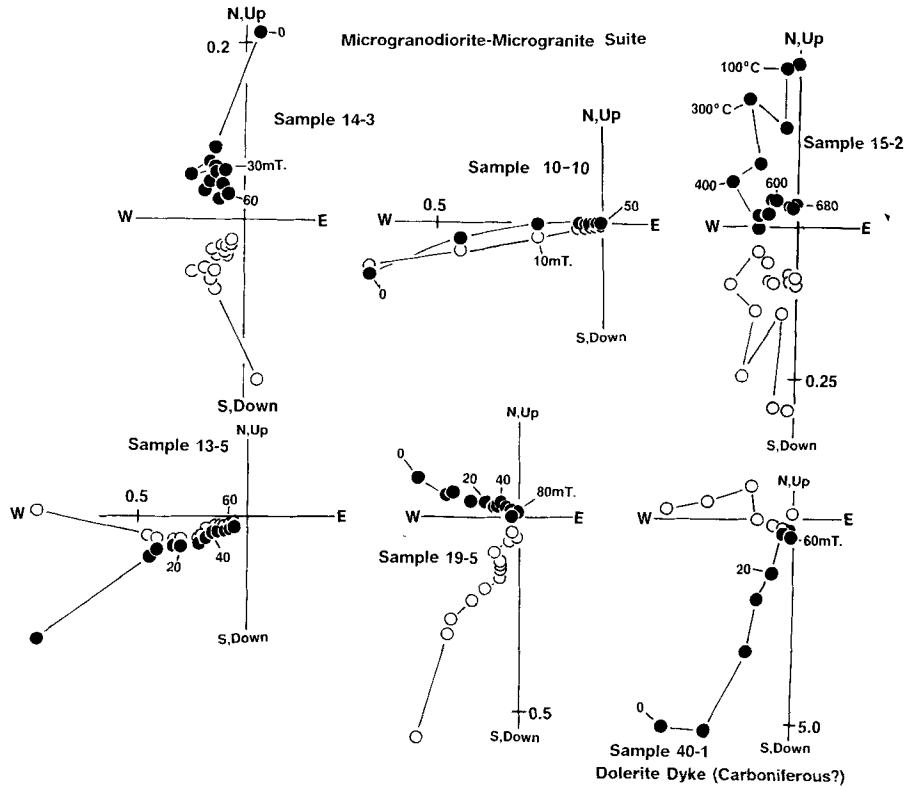


Figure 14. Orthogonal plots illustrating behaviours of members of the microgranite and microgranodiorite suite, Killary area, with progressive AF and thermal demagnetization. Symbols are as for Fig. 4.

mid-Carboniferous (Logmor) dyke of Site 40 has a SSW remanence of low inclination (Fig. 14) contrasting with the remanence in the andesites and microgranodiorites; this is close to the predicted palaeofield direction in this area at 300–290 Ma and compatible with an Upper Carboniferous age. Site 10 is a normally magnetized equivalent of this suite in the Killary area. The TD site (9) in the Kilbride also has a post-Caledonian remanence, although the more westerly reversed declination indicates a contrasting age of acquisition to Sites 10 and 40.

Table 6. Palaeomagnetic results from microdiorite and microgranodiorite intrusions, Killary area.

Site No.	N(n)	R	α_{95}	In Situ D	In Situ I	Tilt Corrected D	Tilt Corrected I
3	6(3)	2.92	26	266	-8	265	6
5	8(5)	4.75	20	279	29	298	12
6	7(7)	6.61	26	291	21	296	-8
7	13(6)	5.88	11	26	70	3	23
	13(3)	2.99	4	242	77	333	43
8	10(7)	6.93	7	103	69	12	27
9	8(3)	2.77	26	73	-34	117	-44
	8(3)	2.91	15	106	73	37	38
10*	4(4)	3.78	27	4	1	359	-47
11	6(5)	4.56	27	284	17	298	12
12	7(6)	5.93	8	275	-15	269	2
13	7(2)	1.87	-	267	5	281	28
14	11(5)	4.55	27	296	3	292	-13
	11(6)	5.54	21	267	62	355	29
15	6(6)	5.71	17	252	74	343	48
16	10(9)	8.47	14	272	11	271	-13
17	8(8)	7.16	20	273	11	272	-13
18	15(9)	8.49	14	281	51	317	24
19	12(10)	8.30	23	281	38	307	22
20	13(9)	8.13	18	209	4	220	44
	13(2)	1.98	-	262	46	306	39

FOOTNOTE: Symbols as for Table 1.
* Dolerite dyke.

5 INTERPRETATION

5.1 Palaeomagnetism and docking of the Dalradian metamorphic terrane and South Mayo Trough

The most important constraint to models reconstructing the former locations of these Caledonian terranes is the close correspondence between the Dalradian stratigraphies in western Ireland and the Grampian Highlands of Scotland (Harris *et al.* 1978). The Dalradian Supergroup was influenced by Grampian orogenesis in both Scotland and Ireland (Lambert & McKerrow 1976) constrained by critical radiometric determinations to the interval 514–491 Ma (Rast, Sturt & Harris 1988). No close correlatives of the Dalradian are found in Newfoundland, although it is believed to correspond in part to the Fleur de Lys Group (Kennedy 1979). The maximum degree of sinistral stike slip which can therefore be entertained between the outcrops in Newfoundland and Ireland is 900 km; this is below the limits of palaeomagnetic detection. Hence we expect palaeomagnetic *inclinations* from the Caledonides of western Ireland to correspond with results from the northern side of the Iapetus Suture in Britain and, following Silurian closure (McKerrow 1988), with results from the south side of the suture as well. Tectonic movements accompanying closure and welding are to be anticipated in anomalies in magnetic *declination* (Macdonald 1980).

Critical Ordovician palaeomagnetic results come from the Connemara gabbros in the DMT and the Mweelrae ignimbrites in the SMT. There are few comparative data from the orthotectonic Caledonides in Scotland. Although

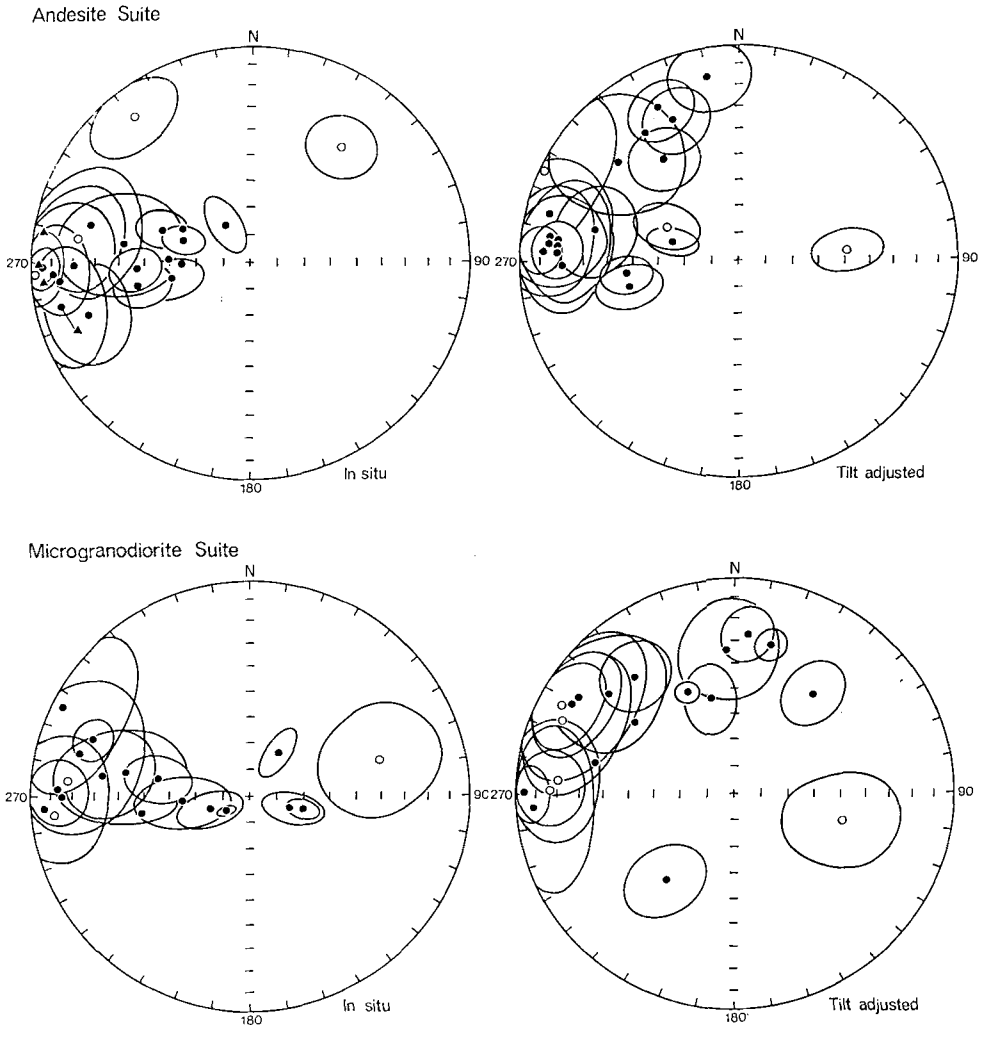


Figure 15. Site mean directions of magnetization derived from minor intrusive suites in the Killary area before and after adjustment for the tilt of the intruded sediments. The triangles are component results from contact samples linked by lines to adjoining intrusions. Pole-centred equal area projections.

the palaeomagnetic record in the Aberdeenshire gabbros (Watts & Briden 1984) has been interpreted as a record of apparent polar wander (APW) during the interval 470–430 Ma (Briden, Turnell & Watts 1984), much of this record is coincident with data from the 'Newer' granites (Torsvik 1985) and intrusive suites within the Northern Highlands (Esang & Piper 1984). Following Torsvik (and see discussion in Section 5.2) it is therefore interpreted as a Silurian imprint acquired during late Caledonian thermochemical activity. Only the remanence in Barrovian zones from the southern Grampian Highlands, which was demonstrably acquired between the D3 and D4 fold generations (Watts 1985, pole BZ in Fig. 16), can be confidently assigned to mid-Ordovician (460–455 Ma) times.

Highest quality results from the paratectonic Caledonides on the south side of the Iapetus suture incorporating both thermal cleaning and palaeomagnetic field tests come from the Builth Volcanic Series of mid-Llanvirn age (Briden & Mullan 1984, pole BuVS in Fig. 16) and from the Borrowdale Volcanic Series of Llandeilo–Caradoc age where the volcanic succession defines a west to east polar migration (author, in preparation, poles BVS1 to BVS3 in Fig. 16). Accordance of remanence declination in the (Llandeilo to early Caradoc) Tramore Volcanics of southern Ireland (Deutsch 1980, pole TR in Fig. 16) with these data confirms the general integrity of SE Ireland with the adjoining paratectonic Caledonides of Britain in mid-Ordovician times.

Following early palaeomagnetic study of the Connemara gabbros from the DMT (Morris *et al.* 1973), Morris & Tanner (1977) evaluated the remanence in the context of the tectonic history and concluded that it was acquired after D3 folding at an intermediate stage in the development of the Connemara antiform (D4). Detailed thermal demagnetization has subsequently resolved three component vectors in the remanence of these bodies (Robertson 1988). Recognizing a component pre-dating latter stages of Caledonian deformation in this area, Robertson (op. cit.) concludes that the palaeopole from this remanence (Table 7) is relatively insensitive to the tectonic model employed. The D4 folding of the Connemara antiform is dated between 460 and 445 Ma (Dempster 1985) and this remanence is therefore

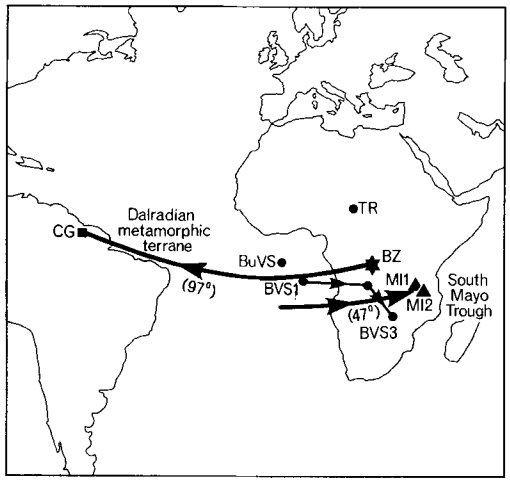


Figure 16. Ordovician palaeomagnetic poles from the Dalradian Metamorphic Terrane (square) and the South Mayo Trough (triangles), Connemara. Comparative poles from the British Caledonides come from the Barrovian zones of the orthotectonic Caledonides (BZ, star) and from the paratectonic Caledonides (circles). The latter are labelled TR (Tramore Volcanics) BuVS (Builth Volcanic Series), BVS1 to BVS3 (Borrowdale Volcanic Series). The arcs are the rotations of the Connemara poles relative to the British reference data inferred to have taken place since the mid-Ordovician ages of magnetization (see text).

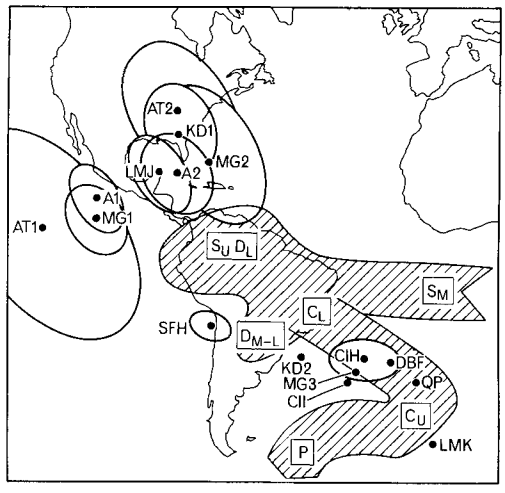


Figure 17. Palaeomagnetic poles from the South Mayo Trough, Connemara, with 95 per cent confidence ovals compared with the contemporaneous south APW path derived from Britain prior to Lower Carboniferous times, and with the collective data from Europe after these times. For a more detailed discussion of this path see Briden *et al.* (1984) and Piper (1987). Representative ages on the path are indicated as S (Silurian), D (Devonian), C (Carboniferous) and P (Permian) with L, M and U referring to the Lower, Middle and Upper divisions of these periods.

essentially contemporaneous with pole BZ from the Barrovian zones in Scotland. The two results are identical following a 97° anticlockwise rotation of Connemara (Fig. 16). This implies that the block incorporating the DMT has been rotated clockwise relative to the Grampian Highlands by this amount since mid-Ordovician (late Llanvirn–Llandeilo) times.

In contrast, the remanence in the older (mid-Llanvirnian) Mweelrae ignimbrites (Deutsch & Somayajulu 1970; Morris *et al.* 1973) in the SMT is rotated anticlockwise from remanence in rocks of contemporaneous age elsewhere (Fig. 16). Apparent rotations are 10° with respect to the orthotectonic Caledonides and 47° with respect to the paratectonic Caledonides. Thus the results summarized in Fig. 16 show that relative motions occurred between the DMT and SMT after mid-Llanvirnian times. Tanner, Dempster & Dickin (1989) interpret movements on the Mannin Thrust dated 447 ± 4 Ma (on the Caradoc–Ashgill boundary) as a record of final docking of these two terranes, while Hutton & Dewey (1986) regard the influx of southerly derived clasts of probable Dalradian lighologies into the Maumtrasna Group (Llanvirn–Llandeilo) in the SMT as the first record of Dalradian molasse in this area. Full welding was complete by the time of deposition of the overstepping Owenduff Group in Lower Silurian times.

5.2 Palaeomagnetism and Late Caledonian (Upper Silurian–Devonian) tectonics of western Ireland

Silurian and younger palaeomagnetic data from Britain define a single apparent polar wander (APW) path compatible with complete closure of the Iapetus suture prior to these times (Fig. 17 and Piper 1987). A long east–west latitudinal APW swathe is representative of the mid-Silurian palaeofield and is recorded notably by the 'Newer' granites in the Scottish Caledonides (Torsvik 1985). Results from this igneous suite are indicative of a widespread thermochemical event extending from the west coast at the Ross of Mull (author, in preparation) to the east coast (Torsvik, op. cit., Esang & Piper 1984) and linked to age dates in the range 430–414 Ma. The same distribution is recorded as an uplift-related imprint in the Scandinavian Caledonides of Central Norway and dated *c.* 420–410 Ma (Piper *et al.* 1990). The equivalent palaeofield inclinations in northern Britain are shallow and in the range 2°–23°. They are distinct from widespread results from the molasse to the orogen and other rocks assigned to late Silurian–Lower Devonian interval (Piper 1987) in which palaeomagnetic inclinations are mostly in the range 40°–55° (Piper 1988) and may briefly have attained even steeper values (Trench *et al.* 1989). Hence rapid (southward) movement of Britain into higher palaeolatitudes during the interval 415–405 Ma is recorded. An accompanying declination change swings the south APW path to the NW (Fig. 17); recent assessments agree that the path of the south pole reached a westerly limit close to 300°E, 5°N during these times (Briden *et al.* 1984; Piper 1987).

A return motion into low latitudes by Middle–Upper Devonian times is indicated by SSW shallow directions of remanence in British rocks of this age. It is poorly constrained by field tests but is not critical to this analysis. A return to intermediate inclinations yielding inclinations similar to the Upper Silurian data is recognized from Lower Carboniferous lavas in Scotland and Derbyshire. The subsequent path illustrated in Fig. 17 is constrained by results from Britain, Eurasia and North America (Piper 1987).

This APW path yields a predictive model for the path of the palaeofield vector at the location of western Ireland as

shown in Fig. 18. It identifies rapid increase in magnetic inclination during mid Silurian to Lower Devonian times followed by a shallowing of the vector later in Devonian times. As a first step to evaluating the palaeomagnetic evidence from the Irish terranes they can be subdivided into results which are highly anomalous in declination with respect to the path derived from Britain (and therefore apparently influenced by prolonged tectonic rotation), and data progressively less divergent in declination from contemporaneous results (and therefore implying smaller amounts of rotation). Whilst interpretation of some results is still open to dispute, this subdivision achieves general conformity with the declining age of the inferred magnetization and forms the basis of Table 7. Stratigraphic ages are related to absolute ages using the time-scale of Harland *et al.* (1982).

The Ordovician (Group A) results clearly merit a tectonic interpretation and have implications discussed above. The Silurian results include data from the Lower–Middle Silurian successions pre-dating Upper Silurian folding (Group B, results AT1 and LMJ in Table 7). Shallow inclination results from the andesite and microgranodiorite suites (A1 and MG1) are compatible with this group; they may fall into this category although fold tests are not conclusive (Table 4). A second set of components (Groups C, KD1, AT2, MG2, A2 in Table 7) have steeper inclinations; they encompass magnetizations post-dating local folding but declinations continue to remain anomalous with respect to the results from Britain. (This conclusion applies whether the results A1 and MG1 in Group B are regarded as pre- or post-folding). Thus the results from western Ireland show the same changes in inclination as the data from Britain but are rotated progressively clockwise with increasing age.

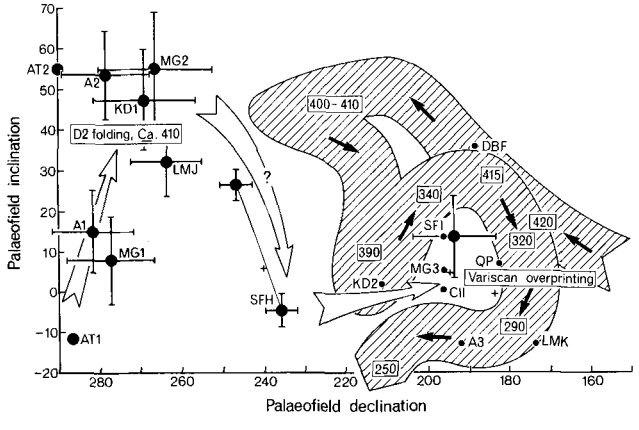


Figure 18. Palaeomagnetic directions (reversed polarity) from the Connemara terranes with 95 per cent confidence limits. They are compared with the path of the palaeofield at this location predicted from the APW path for Britain and Europe as shown in Fig. 17 and represented as a continuous swathe *c.* 10° in width. The figures are representative ages assigned to this path. The location of the D2 folding episode separates palaeomagnetic results showing an improvement in precision with correction for Siluro-Devonian folding from those showing a deterioration in precision when corrected for this deformation. Note that the results from Connemara are significantly removed in declination in a clockwise sense from the results from Britain by an amount which gradually diminishes from mid-Silurian to Carboniferous times. The keys to the poles are given in Table 7 and the crosses refer to results from late Caledonian or younger dolerite dykes of this study.

The succeeding Group D components from Silurian sediments approach the predicted palaeofield path and the case for a tectonic interpretation is thereby diminished. In an earlier study of the Salrock Formation, Morris *et al.* (1973) and Morris (1973) found indications of a remanence related to dewatering or diagenesis, and yielding a bimodal distribution of poles falling within Upper Silurian–Lower Devonian poles from Britain (see Fig. 17). However, the component structure of this remanence was not resolved and the main effect of comprehensive thermal cleaning (Smethurst & Briden 1988) has been to subtract a present field component and shift the remanence vector to a more negative inclination ($D/I = 45°/1°$ to $56°/-5°$). Smethurst & Briden interpret this characteristic ('H') remanence in the Salrock Formation as pre-folding and evidence that local rotation had ceased by mid-Silurian times; this case rests on similarity of the tilt-adjusted inclination to the Clare Island result (CIH in Table 7) which exhibits marked improvement in precision when adjusted for tilt.

However, Clare Island is located within a complex fault zone and this agreement is likely to be fortuitous. The case for a pre-folding remanence in the Salrock Formation is otherwise difficult to sustain for five reasons. (i) The overall precision deteriorates slightly when adjustment for tilt is made. (ii) Fold and contact tests support the view that the contrasting remanence in the intrusions is a thermal or thermochemical phenomenon acquired during initial cooling. (iii) The remanences in the intrusions are mainly of opposite polarity to the host sediments, are at least in part of post-folding origin, and (unlike the sediments) have more marked declination contrast with the British data (Fig. 18). (iv) The intrusions were emplaced before the sediments were fully dewatered, and therefore probably before the magnetic imprint had been fixed in these rocks; in view of the complex origins of magnetic remanence in sediments (Turner 1980; Piper 1987) it seems generally less likely that they will have retained a magnetization pre-dating the remanence in contemporaneous igneous rocks. (v) The aberrant magnetizations identified by Morris (1973, 1976) and confirmed by this study cannot then be readily explained.

Whilst the shallow to intermediate increase in inclination of the Siluro-Devonian palaeofield identified elsewhere in the Caledonides is replicated in the western Ireland data of this age (Fig. 18), the results from this location are rotated systematically clockwise. A declination difference of *c.* 100° in Lower–Middle Silurian times is progressively reduced to 40°–50° after the timing of the local D2 folding. Since axial trends of these folds are now approximately W–E in Connemara (Fig. 1), they were evidently formed in parallelism with contemporaneous folds along the NE strike continuation of the Caledonides, and presumably by the same prevailing stress field. The magnetic inclination of the Salrock Formation records a movement of Connemara back into Lower palaeolatitudes compatible with the Middle–Upper Devonian path from Britain, but a declination anomaly of 20°–30° persists. This conclusion applies whether the *in situ* or tilt adjusted remanence in this formation is accepted (Fig. 18). The Clare Island H component (CIH) is apparently of the same age as the

Table 7. Palaeomagnetic results from the Caledonian terranes of Connemara, western Ireland with the magnetizations organized in order of probable age as defined by field tests and correspondence with the British and Eurasian palaeofield paths.

Code	Rock Unit	Geological age: Stratigraphic	Geological age: Ma	D	I	α_{95}	Pole Position		dp/dm(°)	Notes	Reference
(i)	**Metamorphic Sector, Connemara antiform**										
CG	Connemara gabbros		460-445	238	48	16	302°E	6°N		Remanence acquired after D3 folding and at an intermediate stage of development of the D4 Connemara Antiform	Robertson (1988)
(ii)	**South Mayo Trough**										
	A. Ordovician components predating U.Ordovician - L.Silurian Folding										
MI1	Mweelrae ignimbrites,	O.Llanvirn	473	134	32	20	34°E	9°S	12/22	Remanence acquired before L.Silurian times	Deutsch & Somayajulu(1970) Morris et al. (1973)
MI2				133	30		36°E	9°S	13/24		
	B. Silurian components probably predating Siluro-Devonian folding										
AT1	Albite trachyte lavas, Lough Mask Formation	S,U.Llandovery	430	287	-12	31	243°E	5°N	18/37	Dipolar axis, but only identified at three sites	This study
LMJ	Red sediments, Lough Mask Formation, 'J'	S,U.Llandovery	430	263	32	8	273°E	21°N	5/9	Remanence acquired before Siluro-Devonian folding	"
MG1	Microgranites, W shallow group	S, < Wenlock	425	278	8	11	256°E	8°N	6/12	Emplaced into soft sediments, fold test not diagnostic	This study, see also Morris (1973, 1976).
A1	Andesites, W.shallow group	S, < Wenlock	425	282	14	10	255°E	13°N	6/11	(As above)	"
	C. Components post-dating Siluro-Devonian folding, but pre-dating post-Devonian field and divergent (>30°) from palaeofield path for Britain										
KD1	Dolerites, Kilbride	< S, Wenlock		269	47	12	279°E	22°N	15/24	Emplaced during or after Siluro-Devonian folding	This study
AT2	Albite trachyte lavas, W + overprint	"		290	55	18	270°E	39°N	17/23	Post-folding components	"
MG2	Microgranites, W+ group	"		266	55	14	286°E	26°N	14/20	Post-folding components	This study, see also Morris (1973, 1976)
A2	Andesites, W+ group			278	54	11	277°E	32°N	11/16	Post-folding components	"
	D. Components divergent (<30°) from the palaeofield path for Britain										
SFH	Salrock Formation, 'H' component.	S,Wenlock	425	236	-5	4	287°E	22°S	3/4	Interpreted here as as post- Siluro-Devonian folding	Smethurst & Briden (1988)
CIH	Clare Island redbeds, 'H' component	S,Wenlock?	425	194	14	10	334°E	28°S	5/10	Pre Siluro-Devonian folding, probably rotated within Clew Bay Fault Zone	"
	E. Components correlating with the (post-Devonian) Upper Palaeozoic field for Eurasia (? Variscan overprints)										
LMK	Red sediments, Lough Mask Formation, 'K'overprint	<S.Wenlock		173	-14	11	0°E	43°S	6/11	Post-folding	This study
DBF	Derry Bay Felsite	O or D		189	35	35	341°E	28°S		Probably intrusive into Silurian sediments, no diagnostic tilt test, in situ result used	Morris (1976)
QP	Quartz porphyries, Glensaul-Tourmakeady	O, lower		182	6	4	349°E	31°S	2/4	Post-folding component, interpreted as late Ordovician but correlating with palaeofield axis at 300-290Ma.	Deutsch & Storetvedt (1988)
SFI	Salrock Formation, 'I' overprint			195	14	9	333°E	28°S	5/8	Post-folding component	Smethurst & Briden (1988)
CII	Clare Island redbeds, 'I' overprint			196	0	11	331°E	35°S	6/11	Post-folding component	Smethurst & Briden (1988)
MG3	Microgranodiorites, overprint			196	5	17	331°E	32°S	9/17	Sporadic shallow N-S directed components	This study
A3	Andesites, overprint			192	-13	16	334°E	42°S	8/16	" "	"
KD2	Kilbride dolerites. overprint			211	2	14	314°E	30°S	7/14	" "	"

Group B results but rotated clockwise by only 30° from the palaeofield of this age; this implies that persistent clockwise rotation of Connemara as a whole was largely nullified by anticlockwise block rotation within the Clew Bay Fault Zone (Smethurst & Briden 1988). This conclusion is supported by an early result (but not subject to component analysis), from the Knocknaveen Group (Wenlock) within the landward continuation of this zone in which a remanence with a direction *c.* $D/I = 34°/-54°$ appears to pre-date the D2 folding (Morris *et al.* 1973).

Group E results are sporadic post-folding overprints which plot close to the predicted Lower to Upper Carboniferous palaeofield vector. In the context of the error limits and/or small sample sizes of these data, interpretation of any individual result must be tenuous. However the observation that they tend to plot west of the predicted path is consistent with residual clockwise rotation persisting until mid-Carboniferous times (Fig. 18). Secondary magnetizations in this group were probably imparted by local reheating and fluid migration during the Variscan Orogeny. Deutsch & Storetvedt (1988) find components with this direction resident in a haematite phase in the Glensaul Felsite and Smethurst & Briden (1988) report this component in a wide range of blocking temperatures within the Salrock Formation; it is also present as generally the highest blocking temperature haematite-resident component in the Lough Mask redbeds of this study and is recorded in K–Ar data from dyke swarms (Mitchell & Mohr 1987).

The hypothesis of large and persistent anticlockwise rotation of the western Ireland terranes during Silurian and Devonian times is relevant to the Ordovician rotations summarized in Fig. 16 and discussed above. Ordovician rotation of the DMT is, in fact, equal to the total subsequent rotation (Fig. 18) and therefore records the integrated effect of the Siluro-Devonian tectonic regime. The Ordovician Group A poles from the SMT are probably *c.* 10 Myr older than the pole CG from the Connemara gabbros. Hence either a large counter-clockwise rotation was cancelled by the Siluro-Devonian rotation or, if the antipole to the one plotted in Fig. 16 is relevant to this comparison, a total clockwise rotation of 133° comprises both the Siluro-Devonian and Ordovician contributions. In either case, these mid-Llanvirnian poles must record a component of rotation dating from initial docking of the DMT and the SMT in addition to the integrated effect of the Siluro-Devonian rotation.

The earliest (D1) deformation of Silurian rocks in the SMT is recorded by thrusting and folding. Stretching lineations, deformed conglomerates, tension gash arrays and sinistral extensional shears all collectively indicate that this zone was sited within a sinistral transpressive regime (Hutton & Dewey 1986). The D1 episode was associated with, or merged into, a D2 episode responsible for the upright folds in the study area; it was also associated with sinistral shear. A third deformational episode (D3) was again produced by strike-parallel sinistral shear, which was this time both late and localized. Hutton & Dewey (op. cit.) regard the D1 to D3 episodes as a continuum of sinistral transpressive deformation with the main differences between them relating to the siting of the most intense shear. D1 was focused on the Clew Bay Fault Zone at the northern boundary whereas the D2 episode had its locus along the southern boundary (Doon Rock–Loch Nafooey) fault system (Fig. 2). Collectively these deformations are dated post-mid Wenlock and pre-mid Lower Devonian (Siegenian, see Graham 1981), but smaller sinistral slip and local thrusting continued into Carboniferous times. Currall (1963) and Graham (1981) also describe successive movements influencing Lower and Middle Devonian rocks which include deformation within a sinistral NE–SW directed shear couple (Graham op. cit.).

Thus progressive rotation of the western Ireland terranes during Silurian and Devonian times was accompanied by persistent local and regional sinistral shear along the Scottish-Appalachian sector of the Caledonian Orogen (Pitcher 1969). At a large scale this was produced by impingement of the Avalonian and Cadomian terranes on the Laurentian Foreland (Soper & Hutton 1984). The zone of regional rotation was much more localized: it is recognized by the palaeomagnetic evidence only in Connemara and Mayo although it may extend up to 200 km to the NE where a NE–SW to E–W strike change is recognized as ductile deformation, in part of early Devonian age, in SW Donegal (Rickard 1963).

The geological and palaeomagnetic observations can be explained in terms of large rotations of crustal blocks about vertical axes in zones of distributed continental deformation (McKenzie & Jackson 1986, and Fig. 19). Slip on the fault zones leads to rotation of both the blocks and their boundary faults. This can lead to antithetic rotations of small blocks caught up in these fault zones, as has apparently occurred at Clew Bay. The magnitude of strike slip on the block boundaries is constrained by the width of the blocks and the angle of rotation. Major dislocations in Connemara (Clew Bay, Doon Rock–Lock Nafooey, Galway Bay, Fig. 1) imply block widths of the order of 30–50 km comparable to the dimensions of neotectonic examples illustrated by McKenzie & Jackson (op. cit.). Rotation rates are difficult to estimate because timing of the onset of this regime is unknown. Nevertheless, more than half of the rotation was complete by the D2 folding and initial rotation rates in excess of $3° \text{ Myr}^{-1}$ are required if the regime had commenced by post-Wenlock time (Fig. 19). This rate dropped to less than $1° \text{ Myr}^{-1}$ later in Devonian times as the system of blocks became progressively 'locked' between the suturing boundary plates. Clockwise rotation required by the palaeomagnetic evidence, predicts sinistral strike slip of approximately twice the widths of the blocks if the simple tenets of the model are obeyed. This estimate will be a maximum one because total displacement is accommodated by the range of strain effects distributed across broad zones (Hutton & Dewey 1986), rather than solely by discrete motions along fault planes.

The regional distribution of rotations identified here is probably to be explained in terms of the siting of western Ireland within a 'closure gap' on the Iapetus suture (Phillips *et al.* 1976). At this point a foreland indentor, with a shape inherited from the Laurentian (Precambrian) shield, impinges on the orogen and changes the strike of the suture from NE–SW to approximately E–W between Newfoundland and Ireland (see for example, Dewey 1969; Phillips *et al.* 1976). Following Ordovician suturing along the orogenic belt, this gap would have been closed up by ensuing sinistral strike–slip between the forelands. This explains why

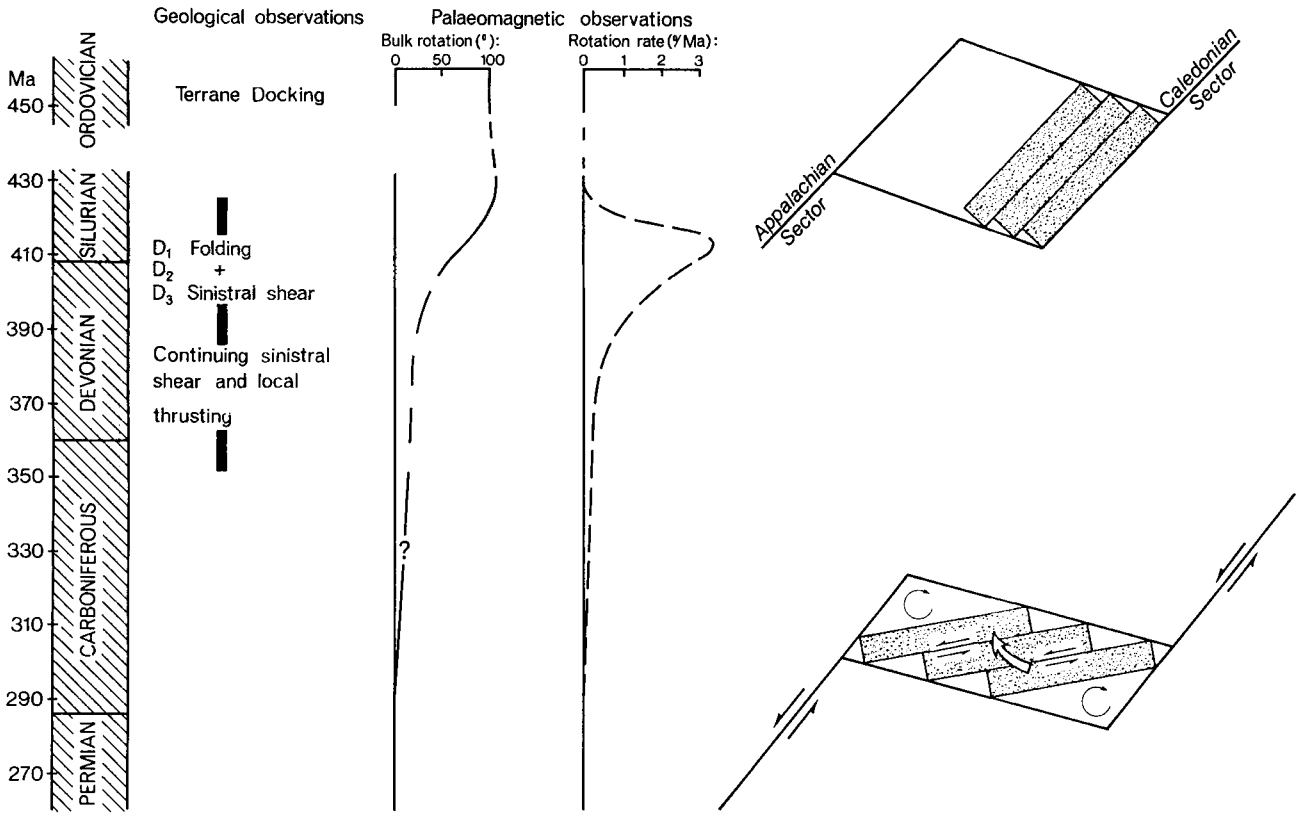


Figure 19. Summary of geological and geophysical evidence relating to the late Caledonian sinistral transpressive regime in western Ireland. The later history of this regime is uncertain in the absence of dated magnetizations after Lower Devonian times and the absolute values of rotation rates assume a mid-Silurian onset for this regime. The sketch is based on the model of McKenzie & Jackson (1986) for block rotation within a zone of distributed deformation. It illustrates in a simplified form how the tectonic blocks in western Ireland were rotated by sinistral transpression across a closure gap in the Caledonian orogenic belt.

distributed deformation in western Ireland had the same life span as the sinistral regime (without rotation) observed elsewhere in the Caledonides (Fig. 19).

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